

Macro Variable Threshold Validation: Testing v2 Results

I ran the full threshold validation experiment across all 12 macro variables plus named combo gates B, F, E, and D. The goal was simple: check whether our current RARE/EXTREME cutoffs in `CONFIG.yaml` still produce the best probability-weighted (PW) forward SPX returns, or whether a different threshold would clear Rohit's four success bars (PW excess +2pp vs current, $n \geq 5$, hit $\geq 60\%$, hostile-regime slice OK).

1. The short version

Phase	What we did	Outcome
P1	Normalized 220 legacy 0–1 percentile rows; fixed <code>per_variable_threshold_sweep.py</code> bands to 0–100	Scale bug fixed; 12/12 vars now have events in corrected sweep
P2	Built and ran <code>threshold_sweep_v2.py</code> on all 12 variables	12 JSONs + <code>SUMMARY.json</code> written
P3	Combo B/F/E/D gate sweeps via leg replay on <code>daily_readings</code>	4 combo JSONs written
P4	This report	See tables below

Bottom line: Most CONFIG thresholds are confirmed at same-side alternatives. **WTI down-side** improves at looser cut (–15% 4wk) but up-side RARE stays weak. Combo B strict 3-of-3 now shows $n=9$ at current gates after fixing HY leg to CONFIG OR (was AND in leg replay).

2. Method, instruments, and test specification

I ran this as a single-variable isolation backtest: for each macro input, I ask whether a different RARE/EXTREME cutoff would produce better probability-weighted (PW) forward SPX returns than the current `CONFIG.yaml` value.

2a. Method

Step	Detail
Event definition	First crossing: variable enters a threshold band from outside the band
Cooldownn	

Step	Detail
	5 calendar days between events (avoids re-counting one prolonged episode)
Forward returns	SPX (<code>^GSPC</code> , Yahoo Finance) measured in NYSE trading days
Horizons tested	1M=21d, 3M=63d, 6M=126d, 9M=189d, 12M=252d
Return framing	Hit rate, avg win, avg loss, PW expected = (hit × avg_win) + ((1-hit) × avg_loss)
Benchmarks (drift)	+0.5% (1M), +2.5% (3M), +5.0% (6M), +7.5% (9M), +10.0% (12M)
Excess	PW expected minus benchmark at that horizon
Hostile slice	Subset where <code>fed_cycle</code> ∈ {HIKING_EARLY, HIKING_LATE, TIGHTENING} OR <code>curve_regime</code> =INVERTED (<code>macro_regime_log</code>)
Verdict methodology	Analyst judgment per variable: combo role, n, hit/avg SPX, economic meaning — not hard Δ/hit gates
Data store	<code>macro_intelligence/data/runic.db</code> → <code>daily_readings</code> (raw_value, unconditional_pctile, meta_json)
Script	<code>scripts/threshold_sweep_v2.py</code> (run date 2026-06-15)
P1 fix	Normalized 220 legacy percentile rows from 0–1 scale to 0–100 before sweep

2b. Duration and range

Item	Value
Event scan start	1990-01-01 (script default <code>--start</code>)
Event scan end	Latest date in <code>daily_readings</code> per variable (mostly 2026-07-03)
SPX history for forwards	Pulled via <code>fetch_yahoo_close('^GSPC')</code> aligned to NYSE session calendar
Primary horizon per variable	VIX/HY/CFTC/NFCI/WALCL/CNH/GSR/VXTS/CPI/CURVE → 3M ; WTI → 6M ; CAPE → 12M
Combo B validated horizon	3M bullish
Combo F validated horizon	6M bullish
Combo E validated horizon	12M bearish
Combo D validated horizon	5D bearish

2c. Band ranges swept (summary)

Variable	Bands tested	Current RARE (CONFIG)	Current EXTREME (CONFIG)
VIX	15, 18, 20, 25, 28, 30, 35, 40 + pctile 65–79	level ≥25 AND pctile ≥80	level ≥35 AND pctile ≥95
HY	300–600 bps + pctile 70/75	OAS ≥400 OR pctile ≥80	OAS ≥500 OR pctile ≥95
CFTC	Short ≤30/20/15/10/5; Long ≥70/80/85/90/95	pctile ≤15 or ≥85	pctile ≤5 or ≥95
NFCI	Easy/tight at ±0.1, ±0.2, ±0.3 , ±0.5, ±0.8 SD	pctile ≤20/≥80 or SD ±0.3	pctile ≤5/≥95 or SD ±0.8
WALCL	Expand/contract at 0.3, 0.5, 0.8 , 1.5, 2.0 , 3.0% MoM	MoM ≥0.8%	MoM ≥2.0%
WTI	Down/up at 3, 5, 6, 8, 10, 15% 4wk	4wk ≥6%	4wk ≥10%
CNH	±0.5, ±1.0, ±1.5 , ±2.5, ±3.5% 4wk	4wk ≥1.5%	4wk ≥3.5%
GSR	Up at 2–10% 4wk	4wk ≥5%	4wk ≥8%
VXTS	Backward ≥1.02–1.20; Contango ≤0.80– 0.95	ratio ≤0.95 or ≥1.10	ratio ≤0.85 or ≥1.20
CAPE	High 22–38; Low 16 , 14, 12	level ≥28 or ≤16	level ≥32 or ≤12
CPI	Hot surprise +0.05 to +0.60 pp	surprise ≥0.2pp	surprise ≥0.4pp
CURVE	Invert –10 to –100 bps; Steepen +5 to +40 bps	spread ≤–30 or steepen ≥15	spread ≤–80 or steepen ≥40

Full per-band hit rates at all five horizons are in the appendix sections below.

3. Infrastructure fixes (P1)

Fix	Before	After
DB percentile normalization	220 rows at 0–1 scale	0 legacy rows remaining; all multiplied by 100
<code>per_variable_threshold_sweep</code> <code>.py</code> bands	0.70–0.79 (0–1 scale), 11 vars	70–79 (0–100 scale), 12 vars incl. CURVE
Corrected regression sweep	VIX high_80_plus n=30 (partial)	VIX high_80_plus n=30; CURVE inverted_80_plus n=7

Legacy rows by variable before migration:

Variable	Legacy rows fixed
VIX	63
NFCI	48

Variable	Legacy rows fixed
GSR	25
WTI	19
CFTC	17
CNH	17
WALCL	15
HY	8
VXTS	8

4. Per-variable: current vs best threshold

Metrics are at each variable's **primary validation horizon** (3M for most; WTI 6M; CAPE 12M). Each variable may have multiple RARE or EXTREME bands (e.g. WTI up/down, CFTC short/long). **Best alt** = best same-side swept alternative at the same tier ($n \geq 5$), ranked by PW excess for bull bands and by bear hit rate then excess for bear bands.

Verdict = analyst judgment (not a hard Δ /hit gate): whether the best alt materially improves CONFIG given the variable's role in combos, sample size, direction (bull/bear), and whether PW excess is economically meaningful (e.g. bearish rows where SPX rallied despite low bear hit).

4a. RARE thresholds (CONFIG vs best RARE alternative)

Hit% ↓ = bearish hit rate (SPX down). **Avg SPX** = mean forward return at primary horizon; for bearish rows, high Avg SPX with 0% hit means the signal missed (SPX rallied). Best alt compares **same-side** bands only (e.g. CAPE high vs high, not high vs low).

CONFIG RARE and **CONFIG EXTREME** columns show the current production cutoffs from `CONFIG.yaml` for that band side (unchanged on Best alt rows so you can compare swept alternatives against what we run today).

Raw returns CSV: `section_4a_rare_threshold_raw_returns.csv` — one row per first-crossing event (4,169 rows) for all RARE-tier and alternative bands (excludes CONFIG EXTREME-only bands). Columns include event date, variable reading, SPX forward returns at 1M/3M/6M/9M/12M, benchmarks, and hostile-regime flags.

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
CAPE	12M	Bear	Current	CAPE_high_28	high ≥ 28	high ≥ 32	7	14.3% ↓	+19.50%	+9.50%	
CAPE	12M	Bear	Best alt	CAPE_high_30	high ≥ 28	high ≥ 32	11	27.3% ↓	+11.71%	+1.71%	Keep CONFIG

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											— Combo E uses CAPE≥2 8; alt high_30 raises bear hit to 27% but still poor timing signal and lowers structural severity
CAPE	12M	Bull	Current	CAPE_low_16	low ≤16	low ≤12	2	100.0%	+18.86%	+8.86%	
CAPE	12M	Bull	Best alt	—	low ≤16	low ≤12	n/a	n/a	n/a	n/a	Keep CONFIG — only 2 deep- value episodes ; threshold defines historic capitulation bucket, not optimizable
CFTC	3M	Bull	Current	CFTC_short_15	pctile ≤15	pctile ≤5	38	63.2%	+1.61%	−0.89%	
CFTC	3M	Bull	Best alt	CFTC_short_10	pctile ≤15	pctile ≤5	30	70.0%	+3.23%	+0.73%	Keep CONFIG — contrarian short-squeeze bucket (63% bull hit);

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											loosenin g to ≤ 10 adds fires with only +1.6pp excess — not worth diluting capitula tion purity
CFTC	3M	Bear	Current	CFTC_lo ng_85	pctile ≥ 85	pctile ≥ 95	39	20.5% ↓	+4.77%	+2.27%	
CFTC	3M	Bear	Best alt	CFTC_lo ng_90	pctile ≥ 85	pctile ≥ 95	35	17.1% ↓	+5.33%	+2.83%	Keep CONFIG — crowde d-fast- money warning for Combo E; bear hit stays ~20% at 3M (structu ral, not timing); long_90 not material ly better
CNH	3M	Bull	Current	CNH_do wn_1.5p ct	4wk $\geq 1.5\%$ (down)	4wk $\geq 3.5\%$ (down)	17	82.4%	+5.08%	+2.58%	
CNH	3M	Bull	Best alt	CNH_do wn_1.0p ct	4wk $\geq 1.5\%$ (down)	4wk $\geq 3.5\%$ (down)	41	85.4%	+4.54%	+2.04%	Keep CONFIG — yuan- strength shock already 82% bull hit at 3M; looser

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											1.0% trades 0.5pp excess for less specific geo signal
CNH	3M	Bear	Current	CNH_up _1.5pct	4wk ≥1.5% (up)	4wk ≥3.5% (up)	22	18.2% ↓	+3.35%	+0.85%	
CNH	3M	Bear	Best alt	CNH_up _0.5pct	4wk ≥1.5% (up)	4wk ≥3.5% (up)	46	30.4% ↓	+2.78%	+0.28%	Keep CONFIG — yuan weakne ss is geo- stress marker but poor SPX down- timing at 3M; no alt fixes economy c story
CPI	3M	Bear	Current	CPI_hot_ 0.20	 surprise ≥0.20pp	 surprise ≥0.40pp	1	0.0% ↓	+5.95%	+3.45%	
CPI	3M	Bear	Best alt	—	 surprise ≥0.20pp	 surprise ≥0.40pp	n/a	n/a	n/a	n/a	Defer — single event at 0.20pp; need more CPI surprise releases before any threshol d move
CURVE	3M	Bear	Current	CURVE_i nvert_30 bps	spread ≤−30bp s	spread ≤−80bp s	10	50.0% ↓	−0.60%	−3.10%	
CURVE	3M	Bear	Best alt				7	57.1% ↓	+0.84%	−1.66%	

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
				CURVE_invert_20bps	spread ≤ -30 bps	spread ≤ -80 bps					Keep CONFIG — -30 bps inversion is standard recession watch; milder -20 bps does not improve recession-timing hit enough to matter
CURVE	3M	Bull	Current	CURVE_steepen_15bps	steepen ≥ 15 bps (4wk)	steepen ≥ 40 bps (4wk)	29	75.9%	+3.53%	+1.03%	
CURVE	3M	Bull	Best alt	CURVE_steepen_5bps	steepen ≥ 15 bps (4wk)	steepen ≥ 40 bps (4wk)	26	65.4%	+1.75%	-0.75%	Keep CONFIG — post-trough steepening (76% bull hit) is recovery confirm; looser 5bps steepen dilutes signal
GSR	3M	Bear	Current	GSR_up_5pct	4wk $\geq 5\%$	4wk $\geq 8\%$	90	24.4% ↓	+3.30%	+0.80%	
GSR	3M	Bear	Best alt	GSR_up_3pct	4wk $\geq 5\%$	4wk $\geq 8\%$	133	34.6% ↓	+1.53%	-0.97%	Keep CONFIG — gold/silver risk-off proxy; 3M SPX timing

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											weak across all bands; 5% is adequate RARE without overfitting
HY	3M	Bear	Current	HY_400 bps	OAS ≥ 400 OR pctile ≥ 80	OAS ≥ 500 OR pctile ≥ 95	42	45.2% ↓	+3.32%	+0.82%	
HY	3M	Bear	Best alt	HY_600 bps	OAS ≥ 400 OR pctile ≥ 80	OAS ≥ 500 OR pctile ≥ 95	25	28.0% ↓	+4.35%	+1.85%	Keep CONFIG — OAS ≥ 400 OR pctile ≥ 800 matches Combo B/A credit stress; tighter 600bps raises excess slightly but cuts n and bear hit
NFCI	3M	Bull	Current	NFCI_easy_0.3	SD ≤ -0.3 OR pctile ≤ 20	SD ≤ -0.8 OR pctile ≤ 5	12	75.0%	+4.42%	+1.92%	
NFCI	3M	Bull	Best alt	NFCI_easy_0.5	SD ≤ -0.3 OR pctile ≤ 20	SD ≤ -0.8 OR pctile ≤ 5	15	66.7%	+2.90%	+0.40%	Keep CONFIG — easy conditions (Combo A/E); small n but 75%

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											hit; alt easy_0.5 loosens definition without economic gain
NFCI	3M	Bear	Current	NFCI_tight_0.3	SD ≥+0.3 OR pctile ≥80	SD ≥+0.8 OR pctile ≥95	8	37.5% ↓	+4.42%	+1.92%	
NFCI	3M	Bear	Best alt	NFCI_tight_0.5	SD ≥+0.3 OR pctile ≥80	SD ≥+0.8 OR pctile ≥95	7	42.9% ↓	−0.01%	−2.51%	Keep CONFIG — tight liquidity marker; n=8 borderline; paired easy/ tight ±0.3 SD is CONFIG standard
VIX	3M	Bear	Current	VIX_25plus	level ≥25 AND pctile ≥80	level ≥35 AND pctile ≥95	72	26.4% ↓	+3.87%	+1.37%	
VIX	3M	Bear	Best alt	VIX_pctile_65_79	level ≥25 AND pctile ≥80	level ≥35 AND pctile ≥95	140	33.6% ↓	+2.27%	−0.23%	Keep CONFIG — n=72 robust; level≥25+pctile ≥80 is Combo B capitulation gate; pctile- only alt trades

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											fear level for weaker discrimination
VXTS	3M	Bear	Current	VXTS_backward_1.10	ratio ≥ 1.10	ratio ≥ 1.20	104	26.0% ↓	+2.21%	−0.29%	
VXTS	3M	Bear	Best alt	VXTS_backward_1.05	ratio ≥ 1.10	ratio ≥ 1.20	85	28.2% ↓	+2.37%	−0.13%	Keep CONFIG — backwardation is stress (Combo D); 3M bear hit ~26% — term structure warns but does not time SPX dips; 1.10 ratio standard
VXTS	3M	Bull	Current	VXTS_contango_0.95	ratio ≤ 0.95	ratio ≤ 0.85	26	76.9%	+4.34%	+1.84%	
VXTS	3M	Bull	Best alt	VXTS_contango_0.90	ratio ≤ 0.95	ratio ≤ 0.85	12	83.3%	+4.59%	+2.09%	Keep CONFIG — complacency (76% bull hit); 0.95 contango is Combo G/D reference; 0.90 alt adds +0.25pp only

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											with n=12
WALCL	3M	Bull	Current	WALCL_expand_0.8	MoM ≥0.8% (expand)	MoM ≥2.0% (expand)	76	68.4%	+2.12%	−0.38%	
WALCL	3M	Bull	Best alt	WALCL_expand_0.5	MoM ≥0.8% (expand)	MoM ≥2.0% (expand)	85	67.1%	+3.04%	+0.54%	Keep CONFIG — QE impulse (Combo A/C); expand_0.5 adds events but excess only +0.9pp; 0.8% MoM is meaningful liquidity injection
WALCL	3M	Bear	Current	WALCL_contract_0.8	MoM ≥0.8% (contract)	MoM ≥2.0% (contract)	66	30.3% ↓	+2.49%	−0.01%	
WALCL	3M	Bear	Best alt	WALCL_contract_1.5	MoM ≥0.8% (contract)	MoM ≥2.0% (contract)	24	29.2% ↓	+3.20%	+0.70%	Keep CONFIG — QT marker; bearish SPX read weak at 3M (liquidity lags); threshold OK
WTI	6M	Bull	Current	WTI_down_6pct	4wk ≥6% (down)	4wk ≥10% (down)	112	67.9%	+4.35%	−0.65%	
WTI	6M	Bull	Best alt				30	73.3%	+5.16%	+0.16%	Consider

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
				WTI_down_15pct	4wk ≥6% (down)	4wk ≥10% (down)					down_15pct — large oil drawdown bucket (n=30, 73% bull hit 6M) better matches supply-shock recovery than ±6% symmetric leg; modest +0.8pp excess vs down_6pct
WTI	6M	Bear	Current	WTI_up_6pct	4wk ≥6% (up)	4wk ≥10% (up)	130	31.5% ↓	+3.01%	−1.99%	
WTI	6M	Bear	Best alt	WTI_up_15pct	4wk ≥6% (up)	4wk ≥10% (up)	40	37.5% ↓	+2.81%	−2.19%	Keep CONFIG — oil spike is stress marker for Combo C but weak SPX-timing (31% bear hit); no same-side alt improves both hit and excess

4b. EXTREME thresholds (CONFIG vs best EXTREME alternative)

CPI / CAPE_low_12: CPI surprise DB starts 2024-01 (31 rows); CAPE ≤12 has 0 first-crossings since 1990.

Raw returns CSV: `section_4b_extreme_threshold_raw_returns.csv` — one row per first-crossing event (3,726 rows) for all EXTREME-tier and alternative bands (excludes CONFIG RARE-only bands). Same column layout as 4a.

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
CAPE	12M	Bear	Current	CAPE_high_32	high ≥28	high ≥32	5	0.0% ↓	+17.84%	+7.84%	
CAPE	12M	Bear	Best alt	CAPE_high_30	high ≥28	high ≥32	11	27.3% ↓	+11.71%	+1.71%	Keep CONFIG — extreme high CAPE is slow-burn valuation risk; alt high_30 is less extreme, not more informative
CAPE	12M	Bull	Current	CAPE_low_12	low ≤16	low ≤12	0	n/a	n/a	n/a	
CAPE	12M	Bull	Best alt	—	low ≤16	low ≤12	n/a	n/a	n/a	n/a	Keep CONFIG — zero first-crossings at ≤12 since 1990; level retained for symmetry with low_16 RARE
CFTC	3M	Bull	Current				18	66.7%	+3.09%	+0.59%	

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
				CFTC_short_5	pctile ≤15	pctile ≤5					
CFTC	3M	Bull	Best alt	CFTC_short_10	pctile ≤15	pctile ≤5	30	70.0%	+3.23%	+0.73%	Keep CONFIG — extreme short positioning aligns with Combo B; alt ≤10 similar profile
CFTC	3M	Bear	Current	CFTC_long_95	pctile ≥85	pctile ≥95	27	18.5% ↓	+5.06%	+2.56%	
CFTC	3M	Bear	Best alt	CFTC_long_90	pctile ≥85	pctile ≥95	35	17.1% ↓	+5.33%	+2.83%	Keep CONFIG — 95th pctile extreme crowding; marginal alt gain not worth retiering
CNH	3M	Bull	Current	CNH_down_3.5pct	4wk ≥1.5% (down)	4wk ≥3.5% (down)	2	50.0%	+0.86%	−1.64%	
CNH	3M	Bull	Best alt	CNH_down_1.0pct	4wk ≥1.5% (down)	4wk ≥3.5% (down)	41	85.4%	+4.54%	+2.04%	Keep CONFIG — only n=2 at extreme; cannot retune; 3.5% remains policy extreme
CNH	3M	Bear	Current	CNH_up_3.5pct	4wk ≥1.5% (up)	4wk ≥3.5% (up)	3	33.3% ↓	+2.77%	+0.27%	

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
CNH	3M	Bear	Best alt	CNH_up_0.5pct	4wk ≥1.5% (up)	4wk ≥3.5% (up)	46	30.4% ↓	+2.78%	+0.28%	Keep CONFIG — n=3 at extreme up; insufficient to move; 3.5% retained
CPI	3M	Bear	Current	CPI_hot_0.40	 surprise ≥0.20pp	 surprise ≥0.40pp	0	n/a	n/a	n/a	
CPI	3M	Bear	Best alt	—	 surprise ≥0.20pp	 surprise ≥0.40pp	n/a	n/a	n/a	n/a	Defer — no EXTREME fires yet; 0.40pp retained pending longer CPI surprise series
CURVE	3M	Bear	Current	CURVE_invert_80bps	spread ≤−30bps	spread ≤−80bps	2	0.0% ↓	+5.01%	+2.51%	
CURVE	3M	Bear	Best alt	CURVE_invert_20bps	spread ≤−30bps	spread ≤−80bps	7	57.1% ↓	+0.84%	−1.66%	Keep CONFIG — deep inversion (n=2) too sparse; −80bps kept as deep recession marker
CURVE	3M	Bull	Current	CURVE_steepen_40bps	steepen ≥15bps (4wk)	steepen ≥40bps (4wk)	17	70.6%	+3.29%	+0.79%	
CURVE	3M	Bull	Best alt				26	65.4%	+1.75%	−0.75%	Keep CONFIG

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
				CURVE_ steepen_ 5bps	steepen ≥15bps (4wk)	steepen ≥40bps (4wk)					— violent steepening already strong (71% hit); looser alts worse
GSR	3M	Bear	Current	GSR_up_ 8pct	4wk ≥5%	4wk ≥8%	39	23.1% ↓	+2.37%	−0.13%	
GSR	3M	Bear	Best alt	GSR_up_ 3pct	4wk ≥5%	4wk ≥8%	133	34.6% ↓	+1.53%	−0.97%	Keep CONFIG — extreme GSR rise is risk- off context, not actionab le SPX timer at 3M
HY	3M	Bear	Current	HY_500 bps	OAS ≥400 OR pctile ≥80	OAS ≥500 OR pctile ≥95	28	35.7% ↓	+1.45%	−1.05%	
HY	3M	Bear	Best alt	HY_600 bps	OAS ≥400 OR pctile ≥80	OAS ≥500 OR pctile ≥95	25	28.0% ↓	+4.35%	+1.85%	Keep CONFIG — 500bps / 95th pctile is standar d extreme widenin g; alt 600bps loses events
VIX	3M	Bear	Current	VIX_35p lus	level ≥25 AND	level ≥35 AND	19	10.5% ↓	+7.91%	+5.41%	

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
					pctile ≥80	pctile ≥95					
VIX	3M	Bear	Best alt	VIX_pctile_65_79	level ≥25 AND pctile ≥80	level ≥35 AND pctile ≥95	140	33.6% ↓	+2.27%	−0.23%	Keep CONFIG — extreme fear episodes (n=19); alt does not improve fear-spike signal meaningfully
VXTS	3M	Bear	Current	VXTS_backward_1.20	ratio ≥1.10	ratio ≥1.20	86	34.9% ↓	+1.61%	−0.89%	
VXTS	3M	Bear	Best alt	VXTS_backward_1.05	ratio ≥1.10	ratio ≥1.20	85	28.2% ↓	+2.37%	−0.13%	Keep CONFIG — extreme backwardation (n=86); marginal alt improvement not worth retiering
VXTS	3M	Bull	Current	VXTS_cointango_0.85	ratio ≤0.95	ratio ≤0.85	4	75.0%	+3.93%	+1.43%	
VXTS	3M	Bull	Best alt	VXTS_cointango_0.90	ratio ≤0.95	ratio ≤0.85	12	83.3%	+4.59%	+2.09%	Keep CONFIG — n=4 at 0.85; too few extreme complacency fires to loosen further

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
WALCL	3M	Bull	Current	WALCL_ expand_ 2.0	MoM ≥0.8% (expand)	MoM ≥2.0% (expand)	43	81.4%	+3.56%	+1.06%	
WALCL	3M	Bull	Best alt	WALCL_ expand_ 0.5	MoM ≥0.8% (expand)	MoM ≥2.0% (expand)	85	67.1%	+3.04%	+0.54%	Keep CONFIG — crisis- era expansi on (81% hit); already strong; no need to retune
WALCL	3M	Bear	Current	WALCL_ contract_ 2.0	MoM ≥0.8% (contract)	MoM ≥2.0% (contract)	11	27.3% ↓	+2.97%	+0.47%	
WALCL	3M	Bear	Best alt	WALCL_ contract_ 1.5	MoM ≥0.8% (contract)	MoM ≥2.0% (contract)	24	29.2% ↓	+3.20%	+0.70%	Keep CONFIG — severe QT episodes rare (n=11); alt does not change policy read
WTI	6M	Bull	Current	WTI_do wn_10p ct	4wk ≥6% (down)	4wk ≥10% (down)	70	68.6%	+3.52%	−1.48%	
WTI	6M	Bull	Best alt	WTI_do wn_15p ct	4wk ≥6% (down)	4wk ≥10% (down)	30	73.3%	+5.16%	+0.16%	Conside r down_1 5pct — same recover y thesis as RARE down leg; +1.6pp excess with

Variable	Hz	Dir	Role	Band	CONFIG RARE	CONFIG EXTREME	n	Hit%	Avg SPX	PW excess	Verdict
											similar hit
WTI	6M	Bear	Current	WTI_up_10pct	4wk ≥6% (up)	4wk ≥10% (up)	93	33.3% ↓	+1.87%	−3.13%	
WTI	6M	Bear	Best alt	WTI_up_15pct	4wk ≥6% (up)	4wk ≥10% (up)	40	37.5% ↓	+2.81%	−2.19%	Keep CONFIG — extreme up-oil fires are real (n=93) but SPX path is mixed; no compelling same-side retune

On the flip side, VIX RARE at ≥ 25 already looks reasonable: 72 events, hostile excess +2.86% at 3M. Tightening to ≥ 30 would drop n to 44 with only +0.66pp more excess, so I would leave VIX RARE at 25.

5. Named combo gate results

5a. Combo B (Capitulation): 3M horizon, bullish

Leg replay on `daily_readings`. **HY leg uses CONFIG OR** ($OAS \geq \text{threshold}$ **or** $\text{pctile} \geq 80$); VIX requires level **and** $\text{pctile} \geq 80$. Rows below are strict **3-of-3** unless noted.

Test	Gate change	n	Hit%	PW excess 3M
CB_VIX_20	VIX ≥ 20 (+pctile ≥ 80)	10	70.0%	+2.92%
CB_VIX_25 (current)	VIX ≥ 25 (+pctile ≥ 80)	9	77.8%	+3.96%
CB_VIX_30	VIX ≥ 30 (+pctile ≥ 80)	7	85.7%	+7.02%
CB_HY_350	HY ≥ 350 bps (OAS OR pctile ≥ 80)	9	77.8%	+3.96%
CB_HY_400 (current)	HY ≥ 400 bps (OAS OR pctile ≥ 80)	9	77.8%	+3.96%
CB_HY_450		7	85.7%	+5.85%

Test	Gate change	n	Hit%	PW excess 3M
	HY $\geq$ 450bps (OAS OR pctl $\geq$ 80)			
CB_CFTC_10	CFTC $\leq$ 10th	7	71.4%	+2.99%
CB_CFTC_15 (current)	CFTC $\leq$ 15th	9	77.8%	+3.96%
CB_CFTC_20	CFTC $\leq$ 20th	10	70.0%	+3.20%
CB_2of3_legs	2-of-3 legs	41	75.6%	+2.15%

With HY OR per CONFIG, strict **3-of-3** first-crossings are rare but not zero: **n=9** at current VIX/HY/ CFTC gates (77.8% hit, +3.96% PW excess 3M). The 2-of-3 variant adds events (n=41) at similar hit rate. Production `combo_detector` still uses AND for HY level+pctl — align if intentional.

5b. Combo F (Recovery): 6M horizon, bullish

SPX threshold above 50WMA	n	Hit%	PW excess 6M
1%	45	80.0%	+2.41%
2%	45	82.2%	+2.27%
3% (current)	42	85.7%	+3.15%
5%	41	85.4%	+3.50%

Current 3% is already strong. Tightening to 5% adds +0.35pp excess with similar hit rate, but the sample is basically the same size. I would keep 3% unless we want fewer marginal reclaims.

5c. Combo E (Valuation Extreme): 12M horizon, bearish

Gate	Value	n	Hit% (down)	PW excess 12M
CAPE $\geq$ 25	25	18	5.6%	+7.51%
CAPE $\geq$ 28 (current)	28	22	9.1%	+5.96%
CAPE $\geq$ 30	30	31	9.7%	+4.74%
CAPE $\geq$ 32	32	30	6.7%	+4.33%
NFCI $\leq$ -0.2	-0.2	23	8.7%	+5.31%
NFCI $\leq$ -0.3 (current)	-0.3	22	9.1%	+5.96%
CFTC $\geq$ 75	75	27	11.1%	+5.07%
CFTC $\geq$ 80 (current)	80	22	9.1%	+5.96%

Current CAPE $\geq$ 28 looks fine. Loosening to 25 adds events but lowers hit rate. Tightening to 30/32 does not improve PW excess enough to bother.

5d. Combo D (FOMO Top): 5D horizon, bearish

Test	Gate	n	Hit% (down)	PW excess 5D
CD_VXTS_1.05	VXTS≥1.05	32	37.5%	−0.23%
CD_VXTS_1.10 (current)	VXTS≥1.10	31	41.9%	−0.28%
CD_VXTS_1.15	VXTS≥1.15	25	40.0%	−0.48%
CD_CFTC_80	CFTC≥80th	33	39.4%	−0.11%
CD_CFTC_85 (current)	CFTC≥85th	31	41.9%	−0.28%
CD_VIX_15	VIX≤15	24	45.8%	−0.47%
CD_VIX_18 (current)	VIX≤18	31	41.9%	−0.28%
CD_2of3_legs	2-of-3	86	33.7%	−0.23%

Combo D does not clear the hit≥60% bar at any gate setting on 5D. Current gates are as good as any alternative we tested, but the signal is weak overall (all variants negative PW excess vs +0.5% benchmark).

6. What I'd change based on this

Priority	Recommendation	Data backing
1	WTI down-side: consider down_15pct for down-leg only (RARE and EXTREME); large drawdown bucket better matches supply-shock recovery thesis; keep symmetric ±6%/10% on up-leg	down_6pct → down_15pct +0.8pp excess, 73% bull hit; up-side weak SPX-timing
2	Combo B: current 3-of-3 gates validated with n=9 after HY OR fix; consider 2-of-3 WATCH promotion (n=41, +2.15% excess)	CB_VIX_25: n=9, hit=77.8%, excess=+3.96%; CB_2of3: n=41
3	Combo F: keep SPX≥3% above 50WMA (optional tighten to 5% for +0.35pp)	3%: hit=85.7%, excess=+3.15%; 5%: hit=85.4%, excess=+3.50%

Everything else I would leave at current CONFIG values for now. HY, VIX, CURVE, and GSR show higher-excess alternatives on paper, but those alts trade off hit rate, sample size, or combo semantics — not worth retiering (see §4 verdicts).

7. Output artifacts

Artifact	Path
Per-variable sweeps (12)	macro_intelligence/analysis/ regime_v2_experiments/threshold_sweep_v2/ *_sweep.json
Cross-variable summary	

Artifact	Path
	macro_intelligence/analysis/ regime_v2_experiments/threshold_sweep_v2/ SUMMARY.json
Combo sweeps	threshold_sweep_v2/ COMBO_B_gate_sweep.json , COMBO_F_spx_sweep.json , COMBO_E_cape_sweep.json , COMBO_D_gate_sweep.json
Corrected pctl regression	macro_intelligence/analysis/ regime_v2_experiments/ F_per_variable_sweep_v2.json
Scripts	scripts/normalize_pctl_scale.py , scripts/threshold_sweep_v2.py , scripts/ combo_gate_sweep_v2.py , scripts/ export_threshold_raw_returns.py
Section 4a raw returns CSV	testing/macro_th_exp/testingv2/ section_4a_rare_threshold_raw_returns.csv (4,169 events)
Section 4b raw returns CSV	testing/macro_th_exp/testingv2/ section_4b_extreme_threshold_raw_returns.c SV (3,726 events)
Raw returns metadata	testing/macro_th_exp/testingv2/ section_4_raw_returns_meta.json

Run timestamps (UTC approx): P1 migration + F sweep ~23s; P2 full sweep ~227s; P3 combo sweeps ~20s.

8. My doubts and questions

- Combo B 3-of-3 with simultaneous VIX level $\geq$ 25 AND pctl $\geq$ 80 AND HY level $\geq$ 400 AND pctl $\geq$ 80 AND CFTC $\leq$ 15 may be too strict for historical first-crossing replay. Combo leg replay now uses HY OR per CONFIG; strict 3-of-3 yields n=9 at current gates. Production `combo_detector` still uses AND for HY — align if intentional.
- WTI's justified change flips from up-side to down-side rare events. That is a paradigm shift, not a simple threshold tweak. Should ROC rare stay symmetric at |6%| or split into directional tiers?
- CPI has only 1 first-crossing at the 0.20pp RARE band since 1990. Do we have enough CPI surprise history in `daily_readings` , or is the sparse count a data coverage issue?

- Combo D never gets close to 60% hit rate on 5D. Is 5D the right horizon, or is this combo structurally more of a 1–2 week fade signal?
- Hostile-regime slices depend on `macro_regime_log` coverage (1,919 rows). Some CAPE and VXTS events fall outside hostile periods with null hostile excess. How much should we weight those nulls in the four-criteria check?

Appendix: full supporting data

9. Instrument and test window reference

Variable	Instrument	Data source	Field tested	Percentile window	Signal direction	Named combos	DB start	DB end	DB rows
CAPE	Shiller CAPE	CAPE scrape	Level (P/E ratio)	Full expanding from 1881-01-01	High=bearish structural (12M)	E	1990-01-05	2026-07-03	1911
CFTC	CFTC TFF Fast Money net	CFTC TFF (S&P E-mini)	Net contracts, 3yr rolling pctile	Rolling 3y from 2006-01-01	Short side bullish; long side bearish	B, D, E, F	2010-06-18	2026-07-03	844
CNH	USD/CNH	Yahoo USDCNH= X / FRED DEXCHUS	4-week % change	Rolling 3y from 2010-01-01	Up=CNH weakness/ risk-off	A, C, G	2010-01-08	2026-07-03	867
CPI	CPI surprise	BLS/FRED vs consensus	Actual minus consensus (pp)	Rolling 3y	Hot CPI bearish (Combo C)	C	2024-01-12	2026-07-03	31
CURVE	10Y-2Y yield spread	FRED T10Y2Y	Spread (bps) + steepen_4 wk_bps in meta	Full expanding from 1976-01-01	Inversion bearish; steepening post- trough	A, E	1990-01-05	2026-07-03	1911
GSR	Gold/ Silver ratio	Yahoo GC=F / SI=F	4-week % change	Rolling 3y	Rising GSR = risk-off	A	2000-09-01	2026-07-03	1355
HY	HY OAS spread	FRED BAMLH0A OHYM2	Level (bps OAS)	Full expanding from 1996-01-01	Bearish when UP (widening)	A, B, F, G	1997-01-02	2026-07-03	7368
NFCI	Chicago Fed NFCI	FRED NFCI	Level (SD units)	Full expanding from 1973-01-01	Easy=bullish, Tight=bearish	A, E	1990-01-05	2026-07-03	1911
VIX						B, D, G	1990-01-05	2026-07-03	1911

Variable	Instrument	Data source	Field tested	Percentile window	Signal direction	Named combos	DB start	DB end	DB rows
	CBOE VIX spot	Yahoo ^VIX	Level (index points)	Full expanding from 1990-01-01	Bearish when UP (fear spike)				
VXTS	VIX term structure	Yahoo ^VIX3M / ^VIX	Ratio (3M VIX / spot VIX)	Full expanding from 2007-01-01	Backward ation bearish; contango complacency	D, G	2007-01-05	2026-07-03	1024
WALCL	Fed balance sheet	FRED WALCL	MoM % change	Full expanding from 2008-01-01	Expansion bullish, contraction bearish	A, C	2003-01-31	2026-07-03	1229
WTI	WTI crude	Yahoo CL=F (FRED fallback)	4-week % change	Rolling 3y	Context-dependent; large down = recovery catalyst	C	2000-08-25	2026-07-03	1356

Forward return instrument: S&P 500 (^GSPC , Yahoo Finance)

Event detection window: 1990-01-01 through latest reading date (variable-specific DB coverage in table)

Horizons (NYSE trading days): 1M=21, 3M=63, 6M=126, 9M=189, 12M=252

Benchmarks (unconditional drift): +0.5% / +2.5% / +5.0% / +7.5% / +10.0%

10. CAPE: full sweep data

CONFIG RARE: {"high_level": 28, "low_level": 16}

CONFIG EXTREME: {"high_level": 32, "low_level": 12}

Primary horizon: 12M (252d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CAPE_high_22		22	6	0.0%	+16.85%	+6.85pp
CAPE_high_25		25	10	0.0%	+22.95%	+12.95pp
CAPE_high_28_C URRENT_RARE	Y	28	7	14.3%	+19.50%	+9.50pp
CAPE_high_30		30	11	27.3%	+11.71%	+1.71pp
	Y	32	5	0.0%	+17.84%	+7.84pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CAPE_high_32_C URRENT_EXTRE ME						
CAPE_high_35		35	5	0.0%	+19.63%	+9.63pp
CAPE_high_38		38	5	33.3%	+7.69%	-2.31pp
CAPE_low_16_C URRENT_RARE	Y	16	2	100.0%	+18.86%	+8.86pp
CAPE_low_14		14	1	100.0%	+66.60%	+56.60pp
CAPE_low_12_C URRENT_EXTRE ME	Y	12	0	n/a	n/a	n/a

10a. CAPE: all horizons per band

CAPE_high_22 | threshold=22 | direction=UP | bullish=False | events n=6

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	6	0.0%	+0.00%	+3.43%	+3.43%	+0.5%	+2.93pp	+2.47pp
3M (63d)	6	33.3%	-6.27%	+8.85%	+3.81%	+2.5%	+1.31pp	+4.96pp
6M (126d)	6	0.0%	+0.00%	+9.08%	+9.08%	+5.0%	+4.08pp	+8.38pp
9M (189d)	6	16.7%	-4.14%	+15.22%	+11.99%	+7.5%	+4.49pp	+14.75pp
12M (252d)	6	0.0%	+0.00%	+16.85%	+16.85%	+10.0%	+6.85pp	+13.36pp

CAPE_high_25 | threshold=25 | direction=UP | bullish=False | events n=10

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	10	10.0%	-1.17%	+4.34%	+3.79%	+0.5%	+3.29pp	+1.43pp
3M (63d)	10	0.0%	+0.00%	+6.79%	+6.79%	+2.5%	+4.29pp	+3.17pp
6M (126d)	10	0.0%	+0.00%	+12.11%	+12.11%	+5.0%	+7.11pp	+6.80pp
9M (189d)	10	0.0%	+0.00%	+14.67%	+14.67%	+7.5%	+7.17pp	+4.85pp
12M (252d)	10	0.0%	+0.00%	+22.95%	+22.95%	+10.0%	+12.95pp	+12.99pp

CAPE_high_28_CURRENT_RARE CURRENT**** | threshold=28 | direction=UP | bullish=False | events n=7

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	7	14.3%	-1.52%	+2.95%	+2.31%	+0.5%	+1.81pp	+1.89pp
3M (63d)	7	0.0%	+0.00%	+8.11%	+8.11%	+2.5%	+5.61pp	+7.13pp

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
6M (126d)	7	0.0%	+0.00%	+10.62%	+10.62%	+5.0%	+5.62pp	+3.16pp
9M (189d)	7	14.3%	-11.07%	+19.59%	+15.21%	+7.5%	+7.71pp	+9.87pp
12M (252d)	7	14.3%	-26.70%	+27.20%	+19.50%	+10.0%	+9.50pp	+15.13pp

CAPE_high_30 | threshold=30 | direction=UP | bullish=False | events n=11

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	11	27.3%	-2.20%	+3.55%	+1.98%	+0.5%	+1.48pp	+1.39pp
3M (63d)	11	36.4%	-6.89%	+5.97%	+1.29%	+2.5%	-1.21pp	+0.83pp
6M (126d)	11	27.3%	-7.73%	+10.42%	+5.47%	+5.0%	+0.47pp	+4.10pp
9M (189d)	11	27.3%	-12.72%	+15.26%	+7.63%	+7.5%	+0.13pp	+5.50pp
12M (252d)	11	27.3%	-16.87%	+22.43%	+11.71%	+10.0%	+1.71pp	+7.41pp

CAPE_high_32_CURRENT_EXTREME CURRENT**** | threshold=32 | direction=UP | bullish=False | events n=5

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	5	0.0%	+0.00%	+3.29%	+3.29%	+0.5%	+2.79pp	+2.24pp
3M (63d)	5	20.0%	-3.52%	+5.84%	+3.97%	+2.5%	+1.47pp	-0.40pp
6M (126d)	5	20.0%	-3.61%	+8.37%	+5.98%	+5.0%	+0.98pp	-2.11pp
9M (189d)	5	0.0%	+0.00%	+15.08%	+15.08%	+7.5%	+7.58pp	+4.75pp
12M (252d)	5	0.0%	+0.00%	+17.84%	+17.84%	+10.0%	+7.84pp	+3.81pp

CAPE_high_35 | threshold=35 | direction=UP | bullish=False | events n=5

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	5	40.0%	-3.57%	+4.92%	+1.52%	+0.5%	+1.02pp	-7.34pp
3M (63d)	5	0.0%	+0.00%	+6.94%	+6.94%	+2.5%	+4.44pp	-0.12pp
6M (126d)	5	20.0%	-6.96%	+14.99%	+10.60%	+5.0%	+5.60pp	+1.74pp
9M (189d)	5	20.0%	-9.07%	+15.83%	+10.85%	+7.5%	+3.35pp	-16.57pp
12M (252d)	5	0.0%	+0.00%	+19.63%	+19.63%	+10.0%	+9.63pp	+2.50pp

CAPE_high_38 | threshold=38 | direction=UP | bullish=False | events n=5

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	5	40.0%	-3.36%	+7.46%	+3.14%	+0.5%	+2.64pp	-6.98pp
3M (63d)	4	50.0%	-8.55%	+7.28%	-0.63%	+2.5%	-3.13pp	-15.05pp
6M (126d)	4	25.0%	-15.04%	+7.97%	+2.22%	+5.0%	-2.78pp	+2.12pp
9M (189d)	4	25.0%	-12.24%	+14.66%	+7.93%	+7.5%	+0.43pp	+7.46pp
12M (252d)	3	33.3%	-18.96%	+21.02%	+7.69%	+10.0%	-2.31pp	+11.08pp

CAPE_low_16_CURRENT_RARE CURRENT**** | threshold=16 | direction=DOWN | bullish=True
 | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	2	0.0%	+0.00%	-3.81%	-3.81%	+0.5%	-4.31pp	n/a
3M (63d)	2	50.0%	+1.75%	-11.15%	-4.70%	+2.5%	-7.20pp	n/a
6M (126d)	2	50.0%	+15.94%	-2.43%	+6.75%	+5.0%	+1.75pp	n/a
9M (189d)	2	100.0%	+12.07%	+0.00%	+12.07%	+7.5%	+4.57pp	n/a
12M (252d)	2	100.0%	+18.86%	+0.00%	+18.86%	+10.0%	+8.86pp	n/a

CAPE_low_14 | threshold=14 | direction=DOWN | bullish=True | events n=1

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	1	100.0%	+22.26%	+0.00%	+22.26%	+0.5%	+21.76pp	n/a
3M (63d)	1	100.0%	+37.56%	+0.00%	+37.56%	+2.5%	+35.06pp	n/a
6M (126d)	1	100.0%	+46.81%	+0.00%	+46.81%	+5.0%	+41.81pp	n/a
9M (189d)	1	100.0%	+60.95%	+0.00%	+60.95%	+7.5%	+53.45pp	n/a
12M (252d)	1	100.0%	+66.60%	+0.00%	+66.60%	+10.0%	+56.60pp	n/a

11. CFTC: full sweep data

CONFIG RARE: {"low_pctile": 15, "high_pctile": 85}

CONFIG EXTREME: {"low_pctile": 5, "high_pctile": 95}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CFTC_short_30		30	58	73.7%	+2.74%	+0.24pp
CFTC_short_20		20	47	62.2%	+1.92%	-0.58pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CFTC_short_15_	Y	15	40	63.2%	+1.61%	-0.89pp
CURRENT_RARE						
CFTC_short_10		10	31	70.0%	+3.23%	+0.73pp
CFTC_short_5_C	Y	5	19	66.7%	+3.09%	+0.59pp
URRENT_EXTRE						
ME						
CFTC_long_70		70	42	14.6%	+3.37%	+0.87pp
CFTC_long_80		80	40	10.3%	+5.81%	+3.31pp
CFTC_long_85_C	Y	85	40	20.5%	+4.77%	+2.27pp
URRENT_RARE						
CFTC_long_90		90	36	17.1%	+5.33%	+2.83pp
CFTC_long_95_C	Y	95	27	18.5%	+5.06%	+2.56pp
URRENT_EXTRE						
ME						

11a. CFTC: all horizons per band

CFTC_short_30 | threshold=30 | direction=DOWN | bullish=True | events n=58

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	58	79.3%	+3.30%	-3.76%	+1.84%	+0.5%	+1.34pp	-0.53pp
3M (63d)	57	73.7%	+6.30%	-7.21%	+2.74%	+2.5%	+0.24pp	-1.14pp
6M (126d)	57	75.4%	+9.86%	-6.33%	+5.88%	+5.0%	+0.88pp	-0.97pp
9M (189d)	57	82.5%	+13.10%	-5.68%	+9.81%	+7.5%	+2.31pp	-1.45pp
12M (252d)	54	94.4%	+16.52%	-7.23%	+15.20%	+10.0%	+5.20pp	+3.93pp

CFTC_short_20 | threshold=20 | direction=DOWN | bullish=True | events n=47

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	45	71.1%	+2.94%	-5.94%	+0.37%	+0.5%	-0.13pp	-0.82pp
3M (63d)	45	62.2%	+6.67%	-5.91%	+1.92%	+2.5%	-0.58pp	-1.99pp
6M (126d)	44	77.3%	+9.60%	-4.88%	+6.31%	+5.0%	+1.31pp	-0.61pp
9M (189d)	43	88.4%	+12.66%	-8.88%	+10.16%	+7.5%	+2.66pp	-2.89pp
12M (252d)	41	97.6%	+16.98%	-12.59%	+16.26%	+10.0%	+6.26pp	+4.26pp

CFTC_short_15_CURRENT_RARE CURRENT**** | threshold=15 | direction=DOWN |
bullish=True | events n=40

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	38	65.8%	+3.87%	-5.27%	+0.75%	+0.5%	+0.25pp	-0.45pp
3M (63d)	38	63.2%	+6.36%	-6.53%	+1.61%	+2.5%	-0.89pp	-1.17pp
6M (126d)	35	74.3%	+10.37%	-4.64%	+6.51%	+5.0%	+1.51pp	+0.51pp
9M (189d)	34	94.1%	+13.66%	-9.37%	+12.31%	+7.5%	+4.81pp	+2.76pp
12M (252d)	32	96.9%	+19.18%	-12.59%	+18.18%	+10.0%	+8.18pp	+8.26pp

CFTC_short_10 | threshold=10 | direction=DOWN | bullish=True | events n=31

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	30	63.3%	+3.32%	-5.32%	+0.15%	+0.5%	-0.35pp	+0.11pp
3M (63d)	30	70.0%	+6.73%	-4.92%	+3.23%	+2.5%	+0.73pp	+1.57pp
6M (126d)	27	77.8%	+10.97%	-2.30%	+8.02%	+5.0%	+3.02pp	+3.90pp
9M (189d)	25	96.0%	+16.64%	-3.36%	+15.84%	+7.5%	+8.34pp	+8.29pp
12M (252d)	23	100.0%	+21.94%	+0.00%	+21.94%	+10.0%	+11.94pp	+11.85pp

CFTC_short_5_CURRENT_EXTREME CURRENT**** | threshold=5 | direction=DOWN | bullish=True | events n=19

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	18	66.7%	+4.01%	-4.66%	+1.12%	+0.5%	+0.62pp	-0.80pp
3M (63d)	18	66.7%	+6.63%	-4.00%	+3.09%	+2.5%	+0.59pp	-1.23pp
6M (126d)	18	83.3%	+10.54%	-4.23%	+8.08%	+5.0%	+3.08pp	+0.14pp
9M (189d)	18	94.4%	+16.17%	-3.36%	+15.09%	+7.5%	+7.59pp	+4.09pp
12M (252d)	18	100.0%	+20.96%	+0.00%	+20.96%	+10.0%	+10.96pp	+7.63pp

CFTC_long_70 | threshold=70 | direction=UP | bullish=False | events n=42

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	42	42.9%	-3.78%	+3.91%	+0.62%	+0.5%	+0.12pp	-0.03pp
3M (63d)	41	14.6%	-6.14%	+5.00%	+3.37%	+2.5%	+0.87pp	-0.51pp
6M (126d)	41	24.4%	-4.34%	+9.58%	+6.19%	+5.0%	+1.19pp	-0.29pp
9M (189d)	41	22.0%	-7.28%	+13.17%	+8.68%	+7.5%	+1.18pp	+1.47pp
12M (252d)	41	24.4%	-4.71%	+14.21%	+9.60%	+10.0%	-0.40pp	-1.01pp

CFTC_long_80 | threshold=80 | direction=UP | bullish=False | events n=40

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	40	20.0%	-3.91%	+4.02%	+2.43%	+0.5%	+1.93pp	+0.87pp
3M (63d)	39	10.3%	-4.32%	+6.96%	+5.81%	+2.5%	+3.31pp	+0.88pp
6M (126d)	39	10.3%	-2.13%	+10.26%	+8.99%	+5.0%	+3.99pp	+1.52pp
9M (189d)	39	25.6%	-6.60%	+15.34%	+9.71%	+7.5%	+2.21pp	+1.12pp
12M (252d)	39	17.9%	-6.73%	+17.10%	+12.82%	+10.0%	+2.82pp	+1.97pp

CFTC_long_85_CURRENT_RARE CURRENT**** | threshold=85 | direction=UP | bullish=False | events n=40

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	40	32.5%	-3.21%	+3.68%	+1.44%	+0.5%	+0.94pp	-1.06pp
3M (63d)	39	20.5%	-3.63%	+6.94%	+4.77%	+2.5%	+2.27pp	+0.19pp
6M (126d)	39	17.9%	-5.64%	+9.27%	+6.60%	+5.0%	+1.60pp	-0.08pp
9M (189d)	39	25.6%	-9.06%	+12.60%	+7.05%	+7.5%	-0.45pp	+0.04pp
12M (252d)	39	20.5%	-8.84%	+14.60%	+9.79%	+10.0%	-0.21pp	+0.43pp

CFTC_long_90 | threshold=90 | direction=UP | bullish=False | events n=36

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	36	36.1%	-3.28%	+4.18%	+1.48%	+0.5%	+0.98pp	-0.88pp
3M (63d)	35	17.1%	-3.55%	+7.16%	+5.33%	+2.5%	+2.83pp	+1.08pp
6M (126d)	35	14.3%	-6.72%	+10.06%	+7.66%	+5.0%	+2.66pp	+1.01pp
9M (189d)	35	20.0%	-9.04%	+13.81%	+9.24%	+7.5%	+1.74pp	+0.41pp
12M (252d)	35	17.1%	-9.45%	+16.34%	+11.92%	+10.0%	+1.92pp	+1.63pp

CFTC_long_95_CURRENT_EXTREME CURRENT**** | threshold=95 | direction=UP | bullish=False | events n=27

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	27	33.3%	-2.49%	+4.07%	+1.89%	+0.5%	+1.39pp	-0.56pp
3M (63d)	27	18.5%	-3.24%	+6.94%	+5.06%	+2.5%	+2.56pp	+0.53pp
6M (126d)	27	7.4%	-9.65%	+10.85%	+9.33%	+5.0%	+4.33pp	+2.60pp
9M (189d)	27	22.2%	-6.85%	+14.91%	+10.07%	+7.5%	+2.57pp	-0.72pp
12M (252d)	27	18.5%	-4.98%	+17.42%	+13.27%	+10.0%	+3.27pp	+1.93pp

12. CNH: full sweep data

CONFIG RARE: {"pct_4wk": 1.5}

CONFIG EXTREME: {"pct_4wk": 3.5}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CNH_down_0.5pct		-0.5	60	74.1%	+3.17%	+0.67pp
CNH_down_1.0pct		-1.0	42	85.4%	+4.54%	+2.04pp
CNH_down_1.5pct_CURRENT_RARE	Y	-1.5	17	82.4%	+5.08%	+2.58pp
CNH_down_2.5pct		-2.5	4	50.0%	+0.85%	-1.65pp
CNH_down_3.5pct_CURRENT_EXTREME	Y	-3.5	2	50.0%	+0.86%	-1.64pp
CNH_up_0.5pct		0.5	47	30.4%	+2.78%	+0.28pp
CNH_up_1.0pct		1.0	27	25.9%	+3.06%	+0.56pp
CNH_up_1.5pct_CURRENT_RARE	Y	1.5	22	18.2%	+3.35%	+0.85pp
CNH_up_2.5pct		2.5	10	30.0%	+3.96%	+1.46pp
CNH_up_3.5pct_CURRENT_EXTRME	Y	3.5	3	33.3%	+2.77%	+0.27pp

12a. CNH: all horizons per band

CNH_down_0.5pct | threshold=-0.5 | direction=DOWN | bullish=True | events n=60

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	59	76.3%	+3.18%	-3.79%	+1.53%	+0.5%	+1.03pp	+1.33pp
3M (63d)	58	74.1%	+6.07%	-5.14%	+3.17%	+2.5%	+0.67pp	+1.55pp
6M (126d)	58	84.5%	+9.92%	-6.56%	+7.36%	+5.0%	+2.36pp	+4.66pp
9M (189d)	57	78.9%	+13.62%	-7.27%	+9.22%	+7.5%	+1.72pp	+2.94pp
12M (252d)	56	87.5%	+16.04%	-8.28%	+13.00%	+10.0%	+3.00pp	+5.13pp

CNH_down_1.0pct | threshold=-1.0 | direction=DOWN | bullish=True | events n=42

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	42	71.4%	+3.29%	-3.32%	+1.40%	+0.5%	+0.90pp	+1.39pp
3M (63d)	41	85.4%	+6.45%	-6.63%	+4.54%	+2.5%	+2.04pp	+2.58pp
6M (126d)	39	87.2%	+10.34%	-5.88%	+8.26%	+5.0%	+3.26pp	+3.93pp
9M (189d)	39	87.2%	+14.19%	-6.11%	+11.59%	+7.5%	+4.09pp	+3.73pp
12M (252d)	38	84.2%	+19.80%	-4.49%	+15.96%	+10.0%	+5.96pp	+8.39pp

CNH_down_1.5pct_CURRENT_RARE CURRENT**** | threshold=-1.5 | direction=DOWN |
bullish=True | events n=17

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	17	76.5%	+3.03%	-3.51%	+1.49%	+0.5%	+0.99pp	+1.18pp
3M (63d)	17	82.4%	+7.56%	-6.51%	+5.08%	+2.5%	+2.58pp	+2.23pp
6M (126d)	17	88.2%	+11.53%	-4.40%	+9.66%	+5.0%	+4.66pp	+3.61pp
9M (189d)	17	100.0%	+13.78%	+0.00%	+13.78%	+7.5%	+6.28pp	+4.51pp
12M (252d)	17	76.5%	+20.93%	-2.77%	+15.36%	+10.0%	+5.36pp	+5.76pp

CNH_down_2.5pct | threshold=-2.5 | direction=DOWN | bullish=True | events n=4

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	4	50.0%	+2.88%	-4.36%	-0.74%	+0.5%	-1.24pp	-1.24pp
3M (63d)	4	50.0%	+5.25%	-3.54%	+0.85%	+2.5%	-1.65pp	-1.65pp
6M (126d)	4	75.0%	+9.76%	-0.30%	+7.24%	+5.0%	+2.24pp	+2.24pp
9M (189d)	4	75.0%	+10.01%	-1.48%	+7.14%	+7.5%	-0.36pp	-0.36pp
12M (252d)	4	75.0%	+15.88%	-6.31%	+10.33%	+10.0%	+0.33pp	+0.33pp

CNH_down_3.5pct_CURRENT_EXTREME CURRENT**** | threshold=-3.5 | direction=DOWN |
bullish=True | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	2	50.0%	+3.43%	-5.37%	-0.97%	+0.5%	-1.47pp	-1.47pp
3M (63d)	2	50.0%	+3.81%	-2.10%	+0.86%	+2.5%	-1.64pp	-1.64pp
6M (126d)	2	100.0%	+9.56%	+0.00%	+9.56%	+5.0%	+4.56pp	+4.56pp
9M (189d)	2	100.0%	+9.52%	+0.00%	+9.52%	+7.5%	+2.02pp	+2.02pp
12M (252d)	2	100.0%	+15.34%	+0.00%	+15.34%	+10.0%	+5.34pp	+5.34pp

CNH_up_0.5pct | threshold=0.5 | direction=UP | bullish=False | events n=47

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	47	36.2%	-4.48%	+4.36%	+1.16%	+0.5%	+0.66pp	-0.80pp
3M (63d)	46	30.4%	-7.17%	+7.13%	+2.78%	+2.5%	+0.28pp	-2.45pp
6M (126d)	46	17.4%	-6.70%	+9.55%	+6.73%	+5.0%	+1.73pp	-0.98pp
9M (189d)	46	17.4%	-3.44%	+14.30%	+11.22%	+7.5%	+3.72pp	+0.75pp
12M (252d)	45	11.1%	-6.19%	+16.71%	+14.17%	+10.0%	+4.17pp	+1.33pp

CNH_up_1.0pct | threshold=1.0 | direction=UP | bullish=False | events n=27

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	27	37.0%	-4.68%	+3.28%	+0.33%	+0.5%	-0.17pp	-0.84pp
3M (63d)	27	25.9%	-4.95%	+5.87%	+3.06%	+2.5%	+0.56pp	-0.24pp
6M (126d)	27	22.2%	-7.55%	+9.02%	+5.34%	+5.0%	+0.34pp	-1.67pp
9M (189d)	27	11.1%	-4.44%	+12.39%	+10.52%	+7.5%	+3.02pp	+1.90pp
12M (252d)	27	11.1%	-5.32%	+16.91%	+14.44%	+10.0%	+4.44pp	+2.75pp

CNH_up_1.5pct_CURRENT_RARE CURRENT**** | threshold=1.5 | direction=UP | bullish=False | events n=22

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	22	40.9%	-7.30%	+3.11%	-1.15%	+0.5%	-1.65pp	-1.65pp
3M (63d)	22	18.2%	-6.34%	+5.51%	+3.35%	+2.5%	+0.85pp	+0.84pp
6M (126d)	22	31.8%	-6.00%	+10.62%	+5.33%	+5.0%	+0.33pp	+0.22pp
9M (189d)	22	13.6%	-4.44%	+12.28%	+10.00%	+7.5%	+2.50pp	+2.94pp
12M (252d)	22	9.1%	-3.54%	+18.25%	+16.27%	+10.0%	+6.27pp	+5.34pp

CNH_up_2.5pct | threshold=2.5 | direction=UP | bullish=False | events n=10

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	10	20.0%	-8.31%	+2.61%	+0.43%	+0.5%	-0.07pp	-0.01pp
3M (63d)	10	30.0%	-1.68%	+6.38%	+3.96%	+2.5%	+1.46pp	+1.36pp
6M (126d)	10	40.0%	-6.99%	+9.03%	+2.62%	+5.0%	-2.38pp	-3.45pp
9M (189d)	10	20.0%	-1.26%	+10.27%	+7.96%	+7.5%	+0.46pp	+3.26pp
12M (252d)	10	10.0%	-0.30%	+11.97%	+10.74%	+10.0%	+0.74pp	-0.42pp

CNH_up_3.5pct_CURRENT_EXTREME CURRENT**** | threshold=3.5 | direction=UP | bullish=False | events n=3

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	3	0.0%	+0.00%	+2.04%	+2.04%	+0.5%	+1.54pp	+1.54pp
3M (63d)	3	33.3%	-0.32%	+4.32%	+2.77%	+2.5%	+0.27pp	+0.27pp
6M (126d)	3	66.7%	-6.60%	+7.70%	-1.83%	+5.0%	-6.83pp	-6.83pp
9M (189d)	3	33.3%	-1.34%	+11.74%	+7.38%	+7.5%	-0.12pp	-0.12pp
12M (252d)	3	33.3%	-0.30%	+11.84%	+7.79%	+10.0%	-2.21pp	-2.21pp

13. CPI: full sweep data

CONFIG RARE: {"surprise_pp": 0.2}

CONFIG EXTREME: {"surprise_pp": 0.4}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CPI_hot_0.05		0.05	3	0.0%	+5.95%	+3.45pp
CPI_hot_0.10		0.1	2	0.0%	+5.95%	+3.45pp
CPI_hot_0.20_CU RRENT_RARE	Y	0.2	2	0.0%	+5.95%	+3.45pp
CPI_hot_0.30		0.3	1	n/a	n/a	n/a
CPI_hot_0.40_CU RRENT_EXTREME	Y	0.4	0	n/a	n/a	n/a
CPI_hot_0.60		0.6	0	n/a	n/a	n/a

13a. CPI: all horizons per band

CPI_hot_0.05 | threshold=0.05 | direction=UP | bullish=False | events n=3

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	1	0.0%	+0.00%	+3.45%	+3.45%	+0.5%	+2.95pp	+2.95pp
3M (63d)	1	0.0%	+0.00%	+5.95%	+5.95%	+2.5%	+3.45pp	+3.45pp
6M (126d)	1	0.0%	+0.00%	+12.04%	+12.04%	+5.0%	+7.04pp	+7.04pp
9M (189d)	1	0.0%	+0.00%	+17.28%	+17.28%	+7.5%	+9.78pp	+9.78pp
12M (252d)	1	0.0%	+0.00%	+22.21%	+22.21%	+10.0%	+12.21pp	+12.21pp

CPI_hot_0.10 | threshold=0.1 | direction=UP | bullish=False | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	1	0.0%	+0.00%	+3.45%	+3.45%	+0.5%	+2.95pp	+2.95pp
3M (63d)	1	0.0%	+0.00%	+5.95%	+5.95%	+2.5%	+3.45pp	+3.45pp
6M (126d)	1	0.0%	+0.00%	+12.04%	+12.04%	+5.0%	+7.04pp	+7.04pp
9M (189d)	1	0.0%	+0.00%	+17.28%	+17.28%	+7.5%	+9.78pp	+9.78pp
12M (252d)	1	0.0%	+0.00%	+22.21%	+22.21%	+10.0%	+12.21pp	+12.21pp

CPI_hot_0.20_CURRENT_RARE CURRENT**** | threshold=0.2 | direction=UP | bullish=False | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	1	0.0%	+0.00%	+3.45%	+3.45%	+0.5%	+2.95pp	+2.95pp
3M (63d)	1	0.0%	+0.00%	+5.95%	+5.95%	+2.5%	+3.45pp	+3.45pp
6M (126d)	1	0.0%	+0.00%	+12.04%	+12.04%	+5.0%	+7.04pp	+7.04pp
9M (189d)	1	0.0%	+0.00%	+17.28%	+17.28%	+7.5%	+9.78pp	+9.78pp
12M (252d)	1	0.0%	+0.00%	+22.21%	+22.21%	+10.0%	+12.21pp	+12.21pp

CPI_hot_0.30 | threshold=0.3 | direction=UP | bullish=False | events n=1

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
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14. CURVE: full sweep data

CONFIG RARE: {"spread_bps": -30, "steepen_4wk_bps": 15}

CONFIG EXTREME: {"spread_bps": -80, "steepen_4wk_bps": 40}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CURVE_invert_1 0bps		-10	9	22.2%	+2.74%	+0.24pp
CURVE_invert_2 0bps		-20	7	57.1%	+0.84%	-1.66pp
CURVE_invert_3 0bps_CURRENT_ RARE	Y	-30	10	50.0%	-0.60%	-3.10pp
CURVE_invert_5 0bps		-50	6	16.7%	+5.24%	+2.74pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
CURVE_invert_8 0bps_CURRENT_ EXTREME	Y	-80	2	0.0%	+5.01%	+2.51pp
CURVE_invert_1 00bps		-100	1	100.0%	-3.65%	-6.15pp
CURVE_steepen_ 5bps		5	26	65.4%	+1.75%	-0.75pp
CURVE_steepen_ 10bps		10	25	64.0%	+1.01%	-1.49pp
CURVE_steepen_ 15bps_CURRENT_ _RARE	Y	15	29	75.9%	+3.53%	+1.03pp
CURVE_steepen_ 25bps		25	30	53.3%	+0.14%	-2.36pp
CURVE_steepen_ 40bps_CURRENT_ _EXTREME	Y	40	17	70.6%	+3.29%	+0.79pp

14a. CURVE: all horizons per band

CURVE_invert_10bps | threshold=-10 | direction=DOWN | bullish=False | events n=9

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	9	11.1%	-4.83%	+5.12%	+4.01%	+0.5%	+3.51pp	+3.55pp
3M (63d)	9	22.2%	-3.31%	+4.47%	+2.74%	+2.5%	+0.24pp	-0.20pp
6M (126d)	9	11.1%	-0.02%	+8.02%	+7.13%	+5.0%	+2.13pp	+1.66pp
9M (189d)	9	11.1%	-6.69%	+6.48%	+5.01%	+7.5%	-2.49pp	-3.08pp
12M (252d)	9	33.3%	-7.27%	+11.86%	+5.49%	+10.0%	-4.51pp	-5.26pp

CURVE_invert_20bps | threshold=-20 | direction=DOWN | bullish=False | events n=7

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	7	42.9%	-3.39%	+5.31%	+1.58%	+0.5%	+1.08pp	+1.08pp
3M (63d)	7	57.1%	-3.47%	+6.57%	+0.84%	+2.5%	-1.66pp	-1.66pp
6M (126d)	7	28.6%	-9.46%	+8.44%	+3.33%	+5.0%	-1.67pp	-1.67pp
9M (189d)	7	42.9%	-12.23%	+15.87%	+3.83%	+7.5%	-3.67pp	-3.67pp
12M (252d)	7	42.9%	-18.69%	+24.65%	+6.08%	+10.0%	-3.92pp	-3.92pp

CURVE_invert_30bps_CURRENT_RARE CURRENT**** | threshold=-30 | direction=DOWN | bullish=False | events n=10

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	10	50.0%	-5.28%	+2.58%	-1.35%	+0.5%	-1.85pp	-1.85pp
3M (63d)	10	50.0%	-5.57%	+4.36%	-0.60%	+2.5%	-3.10pp	-3.10pp
6M (126d)	10	40.0%	-6.32%	+7.14%	+1.76%	+5.0%	-3.24pp	-3.24pp
9M (189d)	10	40.0%	-11.41%	+14.99%	+4.43%	+7.5%	-3.07pp	-3.07pp
12M (252d)	10	30.0%	-19.83%	+15.60%	+4.97%	+10.0%	-5.03pp	-5.03pp

CURVE_invert_50bps | threshold=-50 | direction=DOWN | bullish=False | events n=6

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	6	16.7%	-6.87%	+2.20%	+0.69%	+0.5%	+0.19pp	+0.19pp
3M (63d)	6	16.7%	-2.69%	+6.82%	+5.24%	+2.5%	+2.74pp	+2.74pp
6M (126d)	6	16.7%	-7.08%	+7.66%	+5.21%	+5.0%	+0.21pp	+0.21pp
9M (189d)	6	16.7%	-14.54%	+16.69%	+11.48%	+7.5%	+3.98pp	+3.98pp
12M (252d)	6	16.7%	-24.98%	+22.11%	+14.26%	+10.0%	+4.26pp	+4.26pp

CURVE_invert_80bps_CURRENT_EXTREME CURRENT**** | threshold=-80 | direction=DOWN | bullish=False | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	2	0.0%	+0.00%	+2.01%	+2.01%	+0.5%	+1.51pp	+1.51pp
3M (63d)	2	0.0%	+0.00%	+5.01%	+5.01%	+2.5%	+2.51pp	+2.51pp
6M (126d)	2	0.0%	+0.00%	+9.13%	+9.13%	+5.0%	+4.13pp	+4.13pp
9M (189d)	2	0.0%	+0.00%	+16.72%	+16.72%	+7.5%	+9.22pp	+9.22pp
12M (252d)	2	0.0%	+0.00%	+25.74%	+25.74%	+10.0%	+15.74pp	+15.74pp

CURVE_invert_100bps | threshold=-100 | direction=DOWN | bullish=False | events n=1

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	1	0.0%	+0.00%	+2.84%	+2.84%	+0.5%	+2.34pp	+2.34pp
3M (63d)	1	100.0%	-3.65%	+0.00%	-3.65%	+2.5%	-6.15pp	-6.15pp
6M (126d)	1	0.0%	+0.00%	+7.18%	+7.18%	+5.0%	+2.18pp	+2.18pp
9M (189d)	1	0.0%	+0.00%	+16.97%	+16.97%	+7.5%	+9.47pp	+9.47pp
12M (252d)	1	0.0%	+0.00%	+23.79%	+23.79%	+10.0%	+13.79pp	+13.79pp

CURVE_steepen_5bps | threshold=5 | direction=UP | bullish=True | events n=26

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	26	57.7%	+4.06%	-4.77%	+0.32%	+0.5%	-0.18pp	+0.03pp
3M (63d)	26	65.4%	+6.38%	-6.99%	+1.75%	+2.5%	-0.75pp	-1.59pp
6M (126d)	26	73.1%	+8.81%	-11.01%	+3.47%	+5.0%	-1.53pp	+0.14pp
9M (189d)	26	65.4%	+13.18%	-7.92%	+5.87%	+7.5%	-1.63pp	-0.62pp
12M (252d)	26	61.5%	+15.88%	-8.77%	+6.40%	+10.0%	-3.60pp	-3.13pp

CURVE_steepen_10bps | threshold=10 | direction=UP | bullish=True | events n=25

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	25	56.0%	+3.52%	-5.42%	-0.42%	+0.5%	-0.92pp	-0.98pp
3M (63d)	25	64.0%	+5.88%	-7.65%	+1.01%	+2.5%	-1.49pp	-2.53pp
6M (126d)	25	64.0%	+7.76%	-10.19%	+1.30%	+5.0%	-3.70pp	-1.60pp
9M (189d)	25	64.0%	+11.85%	-9.66%	+4.11%	+7.5%	-3.39pp	-2.61pp
12M (252d)	25	68.0%	+16.16%	-12.29%	+7.06%	+10.0%	-2.94pp	-2.39pp

CURVE_steepen_15bps_CURRENT_RARE CURRENT**** | threshold=15 | direction=UP | bullish=True | events n=29

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	29	62.1%	+3.39%	-4.70%	+0.32%	+0.5%	-0.18pp	-0.21pp
3M (63d)	29	75.9%	+6.82%	-6.80%	+3.53%	+2.5%	+1.03pp	+0.14pp
6M (126d)	29	62.1%	+10.97%	-7.06%	+4.13%	+5.0%	-0.87pp	-0.37pp
9M (189d)	29	62.1%	+14.81%	-10.14%	+5.35%	+7.5%	-2.15pp	-1.58pp
12M (252d)	29	69.0%	+19.31%	-12.68%	+9.39%	+10.0%	-0.61pp	-1.45pp

CURVE_steepen_25bps | threshold=25 | direction=UP | bullish=True | events n=30

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	30	60.0%	+4.11%	-4.80%	+0.55%	+0.5%	+0.05pp	-0.10pp
3M (63d)	30	53.3%	+7.65%	-8.45%	+0.14%	+2.5%	-2.36pp	-1.33pp
6M (126d)	30	56.7%	+11.61%	-6.76%	+3.65%	+5.0%	-1.35pp	-1.80pp
9M (189d)	30	66.7%	+11.43%	-6.56%	+5.43%	+7.5%	-2.07pp	-4.14pp
12M (252d)	30	76.7%	+14.73%	-11.41%	+8.63%	+10.0%	-1.37pp	-4.40pp

CURVE_steepen_40bps_CURRENT_EXTREME CURRENT**** | threshold=40 | direction=UP | bullish=True | events n=17

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	17	76.5%	+4.36%	-3.83%	+2.43%	+0.5%	+1.93pp	+0.79pp
3M (63d)	17	70.6%	+8.19%	-8.48%	+3.29%	+2.5%	+0.79pp	+2.31pp
6M (126d)	17	52.9%	+17.72%	-9.19%	+5.06%	+5.0%	+0.06pp	+1.57pp
9M (189d)	17	64.7%	+20.79%	-11.57%	+9.37%	+7.5%	+1.87pp	+4.60pp
12M (252d)	17	58.8%	+29.11%	-13.20%	+11.69%	+10.0%	+1.69pp	+6.21pp

15. GSR: full sweep data

CONFIG RARE: {"pct_4wk": 5.0}

CONFIG EXTREME: {"pct_4wk": 8.0}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
GSR_up_2pct		2	147	33.8%	+1.66%	-0.84pp
GSR_up_3pct		3	135	34.6%	+1.53%	-0.97pp
GSR_up_4pct		4	113	27.3%	+2.59%	+0.09pp
GSR_up_5pct_CU RRENT_RARE	Y	5	93	24.4%	+3.30%	+0.80pp
GSR_up_6pct		6	72	24.6%	+2.84%	+0.34pp
GSR_up_8pct_CU RRENT_EXTREME	Y	8	41	23.1%	+2.37%	-0.13pp
GSR_up_10pct		10	21	31.6%	+2.35%	-0.15pp

15a. GSR: all horizons per band

GSR_up_2pct | threshold=2 | direction=UP | bullish=False | events n=147

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	146	37.7%	-3.73%	+3.43%	+0.73%	+0.5%	+0.23pp	+0.42pp
3M (63d)	145	33.8%	-6.31%	+5.72%	+1.66%	+2.5%	-0.84pp	+0.04pp
6M (126d)	144	31.2%	-8.40%	+9.31%	+3.78%	+5.0%	-1.22pp	+1.50pp
9M (189d)	143	26.6%	-11.91%	+13.24%	+6.55%	+7.5%	-0.95pp	+2.43pp
12M (252d)	142	25.4%	-14.95%	+16.34%	+8.41%	+10.0%	-1.59pp	+2.67pp

GSR_up_3pct | threshold=3 | direction=UP | bullish=False | events n=135

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	134	38.1%	-4.32%	+3.31%	+0.40%	+0.5%	-0.10pp	-0.24pp
3M (63d)	133	34.6%	-6.37%	+5.71%	+1.53%	+2.5%	-0.97pp	-0.31pp
6M (126d)	132	27.3%	-7.66%	+9.29%	+4.67%	+5.0%	-0.33pp	+1.26pp
9M (189d)	131	27.5%	-12.09%	+13.72%	+6.63%	+7.5%	-0.87pp	+0.81pp
12M (252d)	130	25.4%	-14.85%	+16.44%	+8.50%	+10.0%	-1.50pp	+2.53pp

GSR_up_4pct | threshold=4 | direction=UP | bullish=False | events n=113

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	111	32.4%	-3.66%	+3.72%	+1.32%	+0.5%	+0.82pp	+0.67pp
3M (63d)	110	27.3%	-6.42%	+5.96%	+2.59%	+2.5%	+0.09pp	+0.30pp
6M (126d)	109	26.6%	-6.34%	+9.88%	+5.57%	+5.0%	+0.57pp	+1.99pp
9M (189d)	108	25.0%	-10.56%	+14.21%	+8.02%	+7.5%	+0.52pp	+1.28pp
12M (252d)	108	23.1%	-13.44%	+16.93%	+9.90%	+10.0%	-0.10pp	+1.57pp

GSR_up_5pct_CURRENT_RARE CURRENT**** | threshold=5 | direction=UP | bullish=False | events n=93

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	91	26.4%	-4.15%	+3.95%	+1.82%	+0.5%	+1.32pp	+1.60pp
3M (63d)	90	24.4%	-6.18%	+6.37%	+3.30%	+2.5%	+0.80pp	+1.44pp
6M (126d)	89	27.0%	-6.53%	+10.39%	+5.83%	+5.0%	+0.83pp	+2.60pp
9M (189d)	89	22.5%	-11.71%	+14.85%	+8.89%	+7.5%	+1.39pp	+3.78pp
12M (252d)	89	22.5%	-14.30%	+18.00%	+10.74%	+10.0%	+0.74pp	+4.17pp

GSR_up_6pct | threshold=6 | direction=UP | bullish=False | events n=72

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	70	30.0%	-3.44%	+3.64%	+1.52%	+0.5%	+1.02pp	+1.03pp
3M (63d)	69	24.6%	-6.92%	+6.03%	+2.84%	+2.5%	+0.34pp	+0.61pp
6M (126d)	68	29.4%	-7.47%	+11.36%	+5.82%	+5.0%	+0.82pp	+2.22pp
9M (189d)	68	19.1%	-13.33%	+14.20%	+8.94%	+7.5%	+1.44pp	+2.51pp
12M (252d)	68	20.6%	-13.25%	+18.35%	+11.84%	+10.0%	+1.84pp	+3.86pp

GSR_up_8pct_CURRENT_EXTREME CURRENT**** | threshold=8 | direction=UP | bullish=False | events n=41

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	40	35.0%	-4.32%	+4.91%	+1.68%	+0.5%	+1.18pp	+1.46pp
3M (63d)	39	23.1%	-12.03%	+6.69%	+2.37%	+2.5%	-0.13pp	+1.77pp
6M (126d)	38	28.9%	-12.68%	+12.33%	+5.09%	+5.0%	+0.09pp	+2.98pp
9M (189d)	38	21.1%	-14.62%	+14.25%	+8.17%	+7.5%	+0.67pp	+2.75pp
12M (252d)	38	18.4%	-15.38%	+17.04%	+11.07%	+10.0%	+1.07pp	+4.01pp

GSR_up_10pct | threshold=10 | direction=UP | bullish=False | events n=21

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	20	40.0%	-4.79%	+6.44%	+1.95%	+0.5%	+1.45pp	-0.40pp
3M (63d)	19	31.6%	-11.37%	+8.68%	+2.35%	+2.5%	-0.15pp	+1.31pp
6M (126d)	18	38.9%	-14.03%	+16.00%	+4.32%	+5.0%	-0.68pp	+0.84pp
9M (189d)	18	16.7%	-23.61%	+15.09%	+8.64%	+7.5%	+1.14pp	-1.00pp
12M (252d)	18	16.7%	-17.74%	+17.27%	+11.43%	+10.0%	+1.43pp	+0.19pp

16. HY: full sweep data

CONFIG RARE: {"abs_bps": 400, "high_pctile": 80}

CONFIG EXTREME: {"abs_bps": 500, "high_pctile": 95}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
HY_300bps		300	73	13.6%	+4.24%	+1.74pp
HY_350bps		350	45	17.8%	+4.86%	+2.36pp
HY_400bps_CUR RENT_RARE	Y	400	42	45.2%	+3.32%	+0.82pp
HY_450bps		450	40	25.0%	+3.98%	+1.48pp
HY_500bps_CUR RENT_EXTREME	Y	500	28	35.7%	+1.45%	-1.05pp
HY_600bps		600	25	28.0%	+4.35%	+1.85pp
HY_pctile_70plu s		70	12	9.1%	+12.33%	+9.83pp
HY_pctile_75plu s		75	10	10.0%	+12.32%	+9.82pp

16a. HY: all horizons per band

HY_300bps | threshold=300 | direction=UP | bullish=False | events n=73

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	71	33.8%	-2.71%	+2.78%	+0.92%	+0.5%	+0.42pp	+0.63pp
3M (63d)	66	13.6%	-2.41%	+5.29%	+4.24%	+2.5%	+1.74pp	+3.00pp
6M (126d)	55	5.5%	-2.39%	+10.61%	+9.90%	+5.0%	+4.90pp	+6.40pp
9M (189d)	47	2.1%	-9.07%	+13.12%	+12.65%	+7.5%	+5.15pp	+4.35pp
12M (252d)	38	13.2%	-5.17%	+13.51%	+11.05%	+10.0%	+1.05pp	+0.22pp

HY_350bps | threshold=350 | direction=UP | bullish=False | events n=45

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	45	31.1%	-3.53%	+3.61%	+1.39%	+0.5%	+0.89pp	+0.39pp
3M (63d)	45	17.8%	-6.38%	+7.29%	+4.86%	+2.5%	+2.36pp	+2.68pp
6M (126d)	44	18.2%	-10.50%	+12.51%	+8.32%	+5.0%	+3.32pp	+5.16pp
9M (189d)	43	23.3%	-11.45%	+15.08%	+8.91%	+7.5%	+1.41pp	+4.57pp
12M (252d)	41	22.0%	-12.49%	+21.49%	+14.03%	+10.0%	+4.03pp	+8.04pp

HY_400bps_CURRENT_RARE CURRENT**** | threshold=400 | direction=UP | bullish=False | events n=42

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	42	21.4%	-3.30%	+4.45%	+2.79%	+0.5%	+2.29pp	+2.15pp
3M (63d)	42	45.2%	-4.63%	+9.89%	+3.32%	+2.5%	+0.82pp	+2.84pp
6M (126d)	42	23.8%	-3.39%	+12.88%	+9.01%	+5.0%	+4.01pp	+6.02pp
9M (189d)	42	11.9%	-7.03%	+16.25%	+13.48%	+7.5%	+5.98pp	+9.47pp
12M (252d)	42	19.0%	-10.61%	+22.62%	+16.29%	+10.0%	+6.29pp	+12.33pp

HY_450bps | threshold=450 | direction=UP | bullish=False | events n=40

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	40	25.0%	-3.76%	+4.10%	+2.14%	+0.5%	+1.64pp	+2.78pp
3M (63d)	40	25.0%	-4.46%	+6.80%	+3.98%	+2.5%	+1.48pp	+2.27pp
6M (126d)	40	30.0%	-4.52%	+10.17%	+5.76%	+5.0%	+0.76pp	+2.54pp
9M (189d)	40	17.5%	-8.97%	+13.60%	+9.65%	+7.5%	+2.15pp	+4.82pp

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
12M (252d)	40	25.0%	-13.99%	+15.96%	+8.47%	+10.0%	-1.53pp	+0.64pp

HY_500bps_CURRENT_EXTREME CURRENT**** | threshold=500 | direction=UP | bullish=False | events n=28

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	28	42.9%	-4.97%	+2.54%	-0.68%	+0.5%	-1.18pp	-1.05pp
3M (63d)	28	35.7%	-7.78%	+6.57%	+1.45%	+2.5%	-1.05pp	-3.95pp
6M (126d)	28	46.4%	-11.89%	+10.20%	-0.06%	+5.0%	-5.06pp	-11.53pp
9M (189d)	28	60.7%	-10.28%	+14.43%	-0.57%	+7.5%	-8.07pp	-13.07pp
12M (252d)	28	50.0%	-20.82%	+15.32%	-2.75%	+10.0%	-12.75pp	-24.56pp

HY_600bps | threshold=600 | direction=UP | bullish=False | events n=25

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	25	36.0%	-3.62%	+4.87%	+1.82%	+0.5%	+1.32pp	n/a
3M (63d)	25	28.0%	-5.75%	+8.28%	+4.35%	+2.5%	+1.85pp	n/a
6M (126d)	25	48.0%	-14.87%	+14.84%	+0.58%	+5.0%	-4.42pp	n/a
9M (189d)	25	48.0%	-19.63%	+17.45%	-0.34%	+7.5%	-7.84pp	n/a
12M (252d)	25	40.0%	-21.62%	+14.93%	+0.31%	+10.0%	-9.69pp	n/a

HY_pctile_70plus | threshold=70 | direction=UP | bullish=False | events n=12

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	12	8.3%	-3.13%	+7.35%	+6.48%	+0.5%	+5.98pp	+3.41pp
3M (63d)	11	9.1%	-0.65%	+13.63%	+12.33%	+2.5%	+9.83pp	+7.68pp
6M (126d)	11	9.1%	-1.70%	+19.96%	+17.99%	+5.0%	+12.99pp	+11.45pp
9M (189d)	11	0.0%	+0.00%	+25.81%	+25.81%	+7.5%	+18.31pp	+17.70pp
12M (252d)	11	0.0%	+0.00%	+30.22%	+30.22%	+10.0%	+20.22pp	+20.57pp

HY_pctile_75plus | threshold=75 | direction=UP | bullish=False | events n=10

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	10	10.0%	-3.13%	+7.21%	+6.18%	+0.5%	+5.68pp	+3.41pp
3M (63d)	10	10.0%	-0.65%	+13.77%	+12.32%	+2.5%	+9.82pp	+7.32pp

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
6M (126d)	10	10.0%	-1.70%	+20.31%	+18.11%	+5.0%	+13.11pp	+11.40pp
9M (189d)	10	0.0%	+0.00%	+26.33%	+26.33%	+7.5%	+18.83pp	+18.47pp
12M (252d)	10	0.0%	+0.00%	+30.13%	+30.13%	+10.0%	+20.13pp	+20.49pp

17. NFCI: full sweep data

CONFIG RARE: {"high_pctile": 80, "low_pctile": 20, "high_sd": 0.3, "low_sd": -0.3}

CONFIG EXTREME: {"high_pctile": 95, "low_pctile": 5, "high_sd": 0.8, "low_sd": -0.8}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
NFCI_easy_0.1		-0.1	11	72.7%	+2.48%	-0.02pp
NFCI_easy_0.2		-0.2	13	53.8%	+1.59%	-0.91pp
NFCI_easy_0.3_C URRENT	Y	-0.3	12	75.0%	+4.42%	+1.92pp
NFCI_easy_0.5		-0.5	17	66.7%	+2.90%	+0.40pp
NFCI_easy_0.8		-0.8	6	66.7%	+0.65%	-1.85pp
NFCI_tight_0.3_C URRENT	Y	0.3	8	37.5%	+4.42%	+1.92pp
NFCI_tight_0.5		0.5	7	42.9%	-0.01%	-2.51pp
NFCI_tight_0.8		0.8	4	50.0%	-5.60%	-8.10pp

17a. NFCI: all horizons per band

NFCI_easy_0.1 | threshold=-0.1 | direction=DOWN | bullish=True | events n=11

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	11	72.7%	+5.12%	-2.69%	+2.99%	+0.5%	+2.49pp	+3.77pp
3M (63d)	11	72.7%	+7.05%	-9.71%	+2.48%	+2.5%	-0.02pp	-3.30pp
6M (126d)	11	63.6%	+11.47%	-5.84%	+5.17%	+5.0%	+0.17pp	-3.10pp
9M (189d)	11	81.8%	+12.50%	-13.74%	+7.73%	+7.5%	+0.23pp	-11.01pp
12M (252d)	11	81.8%	+16.12%	-13.47%	+10.74%	+10.0%	+0.74pp	-16.97pp

NFCI_easy_0.2 | threshold=-0.2 | direction=DOWN | bullish=True | events n=13

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	13	61.5%	+4.27%	-5.19%	+0.63%	+0.5%	+0.13pp	-1.49pp
3M (63d)	13	53.8%	+7.63%	-5.45%	+1.59%	+2.5%	-0.91pp	-5.24pp
6M (126d)	13	69.2%	+9.03%	-9.32%	+3.39%	+5.0%	-1.61pp	-5.87pp
9M (189d)	13	76.9%	+12.65%	-13.68%	+6.57%	+7.5%	-0.93pp	-8.44pp
12M (252d)	13	76.9%	+20.31%	-16.50%	+11.81%	+10.0%	+1.81pp	-7.41pp

NFCI_easy_0.3_CURRENT CURRENT**** | threshold=-0.3 | direction=DOWN | bullish=True | events n=12

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	12	75.0%	+4.89%	-1.35%	+3.33%	+0.5%	+2.83pp	+2.01pp
3M (63d)	12	75.0%	+7.04%	-3.43%	+4.42%	+2.5%	+1.92pp	+1.73pp
6M (126d)	12	83.3%	+12.04%	-8.11%	+8.68%	+5.0%	+3.68pp	+9.04pp
9M (189d)	12	91.7%	+12.37%	-8.83%	+10.61%	+7.5%	+3.11pp	+4.95pp
12M (252d)	12	91.7%	+19.59%	-13.41%	+16.84%	+10.0%	+6.84pp	+12.19pp

NFCI_easy_0.5 | threshold=-0.5 | direction=DOWN | bullish=True | events n=17

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	16	68.8%	+3.77%	-4.32%	+1.24%	+0.5%	+0.74pp	+2.57pp
3M (63d)	15	66.7%	+8.27%	-7.84%	+2.90%	+2.5%	+0.40pp	+4.85pp
6M (126d)	15	86.7%	+12.38%	-18.81%	+8.22%	+5.0%	+3.22pp	+8.01pp
9M (189d)	15	73.3%	+16.01%	-12.24%	+8.48%	+7.5%	+0.98pp	+5.79pp
12M (252d)	14	78.6%	+22.36%	-17.09%	+13.91%	+10.0%	+3.91pp	+9.03pp

NFCI_easy_0.8 | threshold=-0.8 | direction=DOWN | bullish=True | events n=6

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	6	66.7%	+3.55%	-1.66%	+1.81%	+0.5%	+1.31pp	+0.71pp
3M (63d)	6	66.7%	+2.53%	-3.11%	+0.65%	+2.5%	-1.85pp	-1.97pp
6M (126d)	6	83.3%	+4.03%	-5.80%	+2.39%	+5.0%	-2.61pp	-0.77pp
9M (189d)	6	100.0%	+8.34%	+0.00%	+8.34%	+7.5%	+0.84pp	+6.27pp
12M (252d)	6	100.0%	+13.37%	+0.00%	+13.37%	+10.0%	+3.37pp	+10.17pp

NFCI_tight_0.3_CURRENT CURRENT**** | threshold=0.3 | direction=UP | bullish=False | events n=8

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	8	25.0%	-1.30%	+4.47%	+3.03%	+0.5%	+2.53pp	+0.60pp
3M (63d)	8	37.5%	-1.56%	+8.01%	+4.42%	+2.5%	+1.92pp	-0.86pp
6M (126d)	8	12.5%	-3.84%	+10.43%	+8.65%	+5.0%	+3.65pp	-2.26pp
9M (189d)	8	25.0%	-6.08%	+15.49%	+10.10%	+7.5%	+2.60pp	-5.78pp
12M (252d)	8	12.5%	-44.47%	+16.79%	+9.13%	+10.0%	-0.87pp	-7.86pp

NFCI_tight_0.5 | threshold=0.5 | direction=UP | bullish=False | events n=7

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	7	71.4%	-3.55%	+3.24%	-1.61%	+0.5%	-2.11pp	+1.60pp
3M (63d)	7	42.9%	-6.60%	+4.93%	-0.01%	+2.5%	-2.51pp	-1.52pp
6M (126d)	7	42.9%	-16.32%	+5.55%	-3.82%	+5.0%	-8.82pp	-2.37pp
9M (189d)	7	57.1%	-25.17%	+10.03%	-10.08%	+7.5%	-17.58pp	-7.58pp
12M (252d)	7	42.9%	-35.62%	+9.29%	-9.96%	+10.0%	-19.96pp	-7.31pp

NFCI_tight_0.8 | threshold=0.8 | direction=UP | bullish=False | events n=4

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	4	50.0%	-5.31%	+4.98%	-0.17%	+0.5%	-0.67pp	n/a
3M (63d)	4	50.0%	-15.29%	+4.09%	-5.60%	+2.5%	-8.10pp	n/a
6M (126d)	4	50.0%	-24.83%	+3.89%	-10.47%	+5.0%	-15.47pp	n/a
9M (189d)	4	50.0%	-30.50%	+6.89%	-11.81%	+7.5%	-19.31pp	n/a
12M (252d)	4	50.0%	-34.07%	+8.90%	-12.58%	+10.0%	-22.58pp	n/a

18. VIX: full sweep data

CONFIG RARE: {"abs_level": 25, "high_pctile": 80}

CONFIG EXTREME: {"abs_level": 35, "high_pctile": 95}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
VIX_15plus		15	79	29.1%	+3.04%	+0.54pp
VIX_18plus		18	78	28.2%	+3.24%	+0.74pp
VIX_20plus		20	78	28.2%	+3.43%	+0.93pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
VIX_25plus_CUR RENT_RARE	Y	25	72	26.4%	+3.87%	+1.37pp
VIX_28plus		28	54	22.6%	+5.01%	+2.51pp
VIX_30plus		30	39	13.2%	+6.09%	+3.59pp
VIX_35plus_CUR RENT_EXTREME	Y	35	19	10.5%	+7.91%	+5.41pp
VIX_40plus		40	12	25.0%	+7.37%	+4.87pp
VIX_pctile_65_79		65	143	33.6%	+2.27%	-0.23pp

18a. VIX: all horizons per band

VIX_15plus | threshold=15 | direction=UP | bullish=False | events n=79

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	79	39.2%	-5.12%	+4.06%	+0.46%	+0.5%	-0.04pp	+0.80pp
3M (63d)	79	29.1%	-7.59%	+7.41%	+3.04%	+2.5%	+0.54pp	+1.27pp
6M (126d)	78	32.1%	-8.28%	+11.13%	+4.91%	+5.0%	-0.09pp	+1.23pp
9M (189d)	78	29.5%	-15.24%	+15.47%	+6.41%	+7.5%	-1.09pp	+2.41pp
12M (252d)	78	26.9%	-18.89%	+18.30%	+8.29%	+10.0%	-1.71pp	+2.55pp

VIX_18plus | threshold=18 | direction=UP | bullish=False | events n=78

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	78	39.7%	-5.12%	+4.12%	+0.45%	+0.5%	-0.05pp	+0.80pp
3M (63d)	78	28.2%	-7.36%	+7.41%	+3.24%	+2.5%	+0.74pp	+1.27pp
6M (126d)	77	31.2%	-8.24%	+11.13%	+5.09%	+5.0%	+0.09pp	+1.23pp
9M (189d)	77	29.9%	-15.24%	+15.70%	+6.46%	+7.5%	-1.04pp	+2.41pp
12M (252d)	77	27.3%	-18.89%	+18.53%	+8.32%	+10.0%	-1.68pp	+2.55pp

VIX_20plus | threshold=20 | direction=UP | bullish=False | events n=78

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	78	39.7%	-4.98%	+4.24%	+0.57%	+0.5%	+0.07pp	+0.85pp
3M (63d)	78	28.2%	-7.36%	+7.66%	+3.43%	+2.5%	+0.93pp	+1.48pp
6M (126d)	77	31.2%	-8.24%	+11.64%	+5.44%	+5.0%	+0.44pp	+1.38pp
9M (189d)	77	29.9%	-15.24%	+15.77%	+6.51%	+7.5%	-0.99pp	+2.06pp
12M (252d)	77	27.3%	-18.89%	+18.88%	+8.58%	+10.0%	-1.42pp	+2.73pp

VIX_25plus_CURRENT_RARE CURRENT**** | threshold=25 | direction=UP | bullish=False | events n=72

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	72	36.1%	-5.22%	+5.01%	+1.31%	+0.5%	+0.81pp	+1.57pp
3M (63d)	72	26.4%	-8.00%	+8.13%	+3.87%	+2.5%	+1.37pp	+2.86pp
6M (126d)	71	31.0%	-8.95%	+12.49%	+5.85%	+5.0%	+0.85pp	+1.66pp
9M (189d)	71	31.0%	-14.72%	+14.11%	+5.18%	+7.5%	-2.32pp	-1.51pp
12M (252d)	71	29.6%	-18.69%	+17.58%	+6.85%	+10.0%	-3.15pp	-1.92pp

VIX_28plus | threshold=28 | direction=UP | bullish=False | events n=54

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	54	33.3%	-4.01%	+5.92%	+2.61%	+0.5%	+2.11pp	+4.14pp
3M (63d)	53	22.6%	-9.35%	+9.21%	+5.01%	+2.5%	+2.51pp	+6.80pp
6M (126d)	52	28.8%	-8.75%	+15.68%	+8.63%	+5.0%	+3.63pp	+6.18pp
9M (189d)	52	30.8%	-13.89%	+18.64%	+8.63%	+7.5%	+1.13pp	+4.64pp
12M (252d)	52	28.8%	-17.94%	+23.22%	+11.35%	+10.0%	+1.35pp	+6.53pp

VIX_30plus | threshold=30 | direction=UP | bullish=False | events n=39

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	39	30.8%	-5.42%	+6.89%	+3.10%	+0.5%	+2.60pp	+4.60pp
3M (63d)	38	13.2%	-11.01%	+8.68%	+6.09%	+2.5%	+3.59pp	+5.85pp
6M (126d)	38	23.7%	-9.83%	+16.43%	+10.21%	+5.0%	+5.21pp	+5.73pp
9M (189d)	38	23.7%	-13.15%	+19.93%	+12.10%	+7.5%	+4.60pp	+5.02pp
12M (252d)	38	23.7%	-13.96%	+24.60%	+15.46%	+10.0%	+5.46pp	+5.27pp

VIX_35plus_CURRENT_EXTREME CURRENT**** | threshold=35 | direction=UP | bullish=False | events n=19

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	19	31.6%	-7.28%	+7.23%	+2.65%	+0.5%	+2.15pp	+5.91pp
3M (63d)	19	10.5%	-8.55%	+9.85%	+7.91%	+2.5%	+5.41pp	+13.69pp
6M (126d)	19	10.5%	-16.83%	+18.67%	+14.93%	+5.0%	+9.93pp	+19.24pp
9M (189d)	19	10.5%	-18.66%	+21.12%	+16.93%	+7.5%	+9.43pp	+19.73pp
12M (252d)	19	15.8%	-13.98%	+27.16%	+20.66%	+10.0%	+10.66pp	+16.85pp

VIX_40plus | threshold=40 | direction=UP | bullish=False | events n=12

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	12	33.3%	-8.67%	+6.87%	+1.69%	+0.5%	+1.19pp	+5.49pp
3M (63d)	12	25.0%	-6.88%	+12.13%	+7.37%	+2.5%	+4.87pp	+19.16pp
6M (126d)	12	16.7%	-12.22%	+19.79%	+14.45%	+5.0%	+9.45pp	+26.43pp
9M (189d)	12	8.3%	-19.85%	+21.64%	+18.18%	+7.5%	+10.68pp	+29.36pp
12M (252d)	12	16.7%	-9.51%	+27.47%	+21.31%	+10.0%	+11.31pp	+24.39pp

VIX_pctile_65_79 | threshold=65 | direction=UP | bullish=False | events n=143

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	141	33.3%	-4.06%	+4.03%	+1.33%	+0.5%	+0.83pp	+1.13pp
3M (63d)	140	33.6%	-8.12%	+7.53%	+2.27%	+2.5%	-0.23pp	+0.31pp
6M (126d)	139	30.9%	-11.05%	+10.51%	+3.84%	+5.0%	-1.16pp	+0.90pp
9M (189d)	137	28.5%	-16.00%	+13.89%	+5.38%	+7.5%	-2.12pp	+1.38pp
12M (252d)	136	30.9%	-17.41%	+18.30%	+7.27%	+10.0%	-2.73pp	+2.38pp

19. VXTS: full sweep data

CONFIG RARE: {"low_ratio": 0.95, "high_ratio": 1.1}

CONFIG EXTREME: {"low_ratio": 0.85, "high_ratio": 1.2}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
VXTS_backward_1.02		1.02	66	27.0%	+2.52%	+0.02pp
VXTS_backward_1.05		1.05	87	28.2%	+2.37%	-0.13pp
VXTS_backward_1.10_CURRENT_RARE	Y	1.1	106	26.0%	+2.21%	-0.29pp
VXTS_backward_1.15		1.15	118	26.1%	+2.76%	+0.26pp
VXTS_backward_1.20_CURRENT_EXTREME	Y	1.2	88	34.9%	+1.61%	-0.89pp
	Y	0.95	27	76.9%	+4.34%	+1.84pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
VXTS_contango_ 0.95_CURRENT_ RARE						
VXTS_contango_ 0.90		0.9	12	83.3%	+4.59%	+2.09pp
VXTS_contango_ 0.85_CURRENT_ EXTREME	Y	0.85	4	75.0%	+3.93%	+1.43pp
VXTS_contango_ 0.80		0.8	2	50.0%	-0.09%	-2.59pp

19a. VXTS: all horizons per band

VXTS_backward_1.02 | threshold=1.02 | direction=UP | bullish=False | events n=66

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	65	26.2%	-4.38%	+3.47%	+1.42%	+0.5%	+0.92pp	+0.20pp
3M (63d)	63	27.0%	-6.93%	+6.01%	+2.52%	+2.5%	+0.02pp	+1.01pp
6M (126d)	62	25.8%	-8.50%	+9.50%	+4.85%	+5.0%	-0.15pp	+0.25pp
9M (189d)	62	32.3%	-12.46%	+12.84%	+4.68%	+7.5%	-2.82pp	-2.90pp
12M (252d)	62	25.8%	-20.12%	+16.62%	+7.14%	+10.0%	-2.86pp	-3.28pp

VXTS_backward_1.05 | threshold=1.05 | direction=UP | bullish=False | events n=87

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	86	29.1%	-5.20%	+3.22%	+0.78%	+0.5%	+0.28pp	+0.65pp
3M (63d)	85	28.2%	-7.13%	+6.11%	+2.37%	+2.5%	-0.13pp	+1.63pp
6M (126d)	84	28.6%	-10.88%	+10.24%	+4.20%	+5.0%	-0.80pp	+0.66pp
9M (189d)	83	27.7%	-13.01%	+13.20%	+5.94%	+7.5%	-1.56pp	-1.97pp
12M (252d)	82	22.0%	-17.90%	+16.84%	+9.22%	+10.0%	-0.78pp	+0.86pp

VXTS_backward_1.10_CURRENT_RARE CURRENT**** | threshold=1.1 | direction=UP |
bullish=False | events n=106

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	105	32.4%	-4.07%	+3.28%	+0.90%	+0.5%	+0.40pp	-0.06pp
3M (63d)	104	26.0%	-8.20%	+5.85%	+2.21%	+2.5%	-0.29pp	+0.23pp
6M (126d)	103	27.2%	-8.17%	+9.58%	+4.76%	+5.0%	-0.24pp	+0.74pp

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
9M (189d)	101	21.8%	-9.81%	+12.38%	+7.54%	+7.5%	+0.04pp	+0.50pp
12M (252d)	99	16.2%	-13.45%	+15.24%	+10.60%	+10.0%	+0.60pp	+1.33pp

VXTS_backward_1.15 | threshold=1.15 | direction=UP | bullish=False | events n=118

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	117	39.3%	-3.77%	+3.13%	+0.42%	+0.5%	-0.08pp	-0.34pp
3M (63d)	115	26.1%	-5.27%	+5.59%	+2.76%	+2.5%	+0.26pp	+0.37pp
6M (126d)	115	25.2%	-7.48%	+9.41%	+5.15%	+5.0%	+0.15pp	+1.49pp
9M (189d)	113	20.4%	-8.52%	+11.99%	+7.81%	+7.5%	+0.31pp	+1.05pp
12M (252d)	111	15.3%	-11.61%	+14.17%	+10.22%	+10.0%	+0.22pp	+2.25pp

VXTS_backward_1.20_CURRENT_EXTREME CURRENT**** | threshold=1.2 | direction=UP | bullish=False | events n=88

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	86	39.5%	-3.12%	+2.38%	+0.20%	+0.5%	-0.30pp	-0.82pp
3M (63d)	86	34.9%	-4.82%	+5.06%	+1.61%	+2.5%	-0.89pp	-0.76pp
6M (126d)	86	29.1%	-6.56%	+7.65%	+3.52%	+5.0%	-1.48pp	-0.48pp
9M (189d)	84	22.6%	-6.87%	+11.27%	+7.17%	+7.5%	-0.33pp	+1.20pp
12M (252d)	81	21.0%	-7.05%	+13.16%	+8.92%	+10.0%	-1.08pp	+1.64pp

VXTS_contango_0.95_CURRENT_RARE CURRENT**** | threshold=0.95 | direction=DOWN | bullish=True | events n=27

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	27	74.1%	+6.34%	-6.87%	+2.91%	+0.5%	+2.41pp	+3.15pp
3M (63d)	26	76.9%	+9.19%	-11.83%	+4.34%	+2.5%	+1.84pp	+2.80pp
6M (126d)	25	68.0%	+15.31%	-12.06%	+6.55%	+5.0%	+1.55pp	+4.48pp
9M (189d)	25	68.0%	+19.30%	-19.00%	+7.05%	+7.5%	-0.45pp	+1.57pp
12M (252d)	25	68.0%	+22.56%	-23.76%	+7.74%	+10.0%	-2.26pp	+2.27pp

VXTS_contango_0.90 | threshold=0.9 | direction=DOWN | bullish=True | events n=12

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	12	66.7%	+6.61%	-9.60%	+1.21%	+0.5%	+0.71pp	+4.65pp
3M (63d)	12	83.3%	+9.43%	-19.59%	+4.59%	+2.5%	+2.09pp	+5.69pp
6M (126d)	12	75.0%	+16.63%	-14.72%	+8.80%	+5.0%	+3.80pp	+2.23pp
9M (189d)	12	83.3%	+19.37%	-16.57%	+13.38%	+7.5%	+5.88pp	+1.04pp
12M (252d)	12	75.0%	+27.20%	-11.53%	+17.52%	+10.0%	+7.52pp	-2.40pp

VXTS_contango_0.85_CURRENT_EXTREME CURRENT**** | threshold=0.85 | direction=DOWN
| bullish=True | events n=4

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	4	25.0%	+10.50%	-8.20%	-3.53%	+0.5%	-4.03pp	n/a
3M (63d)	4	75.0%	+10.45%	-15.63%	+3.93%	+2.5%	+1.43pp	n/a
6M (126d)	4	50.0%	+25.39%	-13.26%	+6.07%	+5.0%	+1.07pp	n/a
9M (189d)	4	75.0%	+21.22%	-19.85%	+10.95%	+7.5%	+3.45pp	n/a
12M (252d)	4	75.0%	+25.50%	-5.35%	+17.79%	+10.0%	+7.79pp	n/a

VXTS_contango_0.80 | threshold=0.8 | direction=DOWN | bullish=True | events n=2

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	2	50.0%	+2.22%	-11.09%	-4.43%	+0.5%	-4.93pp	n/a
3M (63d)	2	50.0%	+3.05%	-3.22%	-0.09%	+2.5%	-2.59pp	n/a
6M (126d)	2	50.0%	+17.95%	-6.42%	+5.77%	+5.0%	+0.77pp	n/a
9M (189d)	2	100.0%	+11.80%	+0.00%	+11.80%	+7.5%	+4.30pp	n/a
12M (252d)	2	100.0%	+25.88%	+0.00%	+25.88%	+10.0%	+15.88pp	n/a

20. WALCL: full sweep data

CONFIG RARE: {"mom_pct": 0.8}

CONFIG EXTREME: {"mom_pct": 2.0}

Primary horizon: 3M (63d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
WALCL_expand_0.3		0.3	95	70.2%	+2.78%	+0.28pp
		0.5	86	67.1%	+3.04%	+0.54pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
WALCL_expand_0.5						
WALCL_expand_0.8_CURRENT_RARE	Y	0.8	76	68.4%	+2.12%	-0.38pp
WALCL_expand_1.5		1.5	52	71.2%	+1.99%	-0.51pp
WALCL_expand_2.0_CURRENT_EXTREME	Y	2.0	43	81.4%	+3.56%	+1.06pp
WALCL_expand_3.0		3.0	22	68.2%	+2.87%	+0.37pp
WALCL_contract_0.3		-0.3	89	25.8%	+2.86%	+0.36pp
WALCL_contract_0.5		-0.5	85	27.1%	+2.50%	+0.00pp
WALCL_contract_0.8_CURRENT_RARE	Y	-0.8	66	30.3%	+2.49%	-0.01pp
WALCL_contract_1.5		-1.5	24	29.2%	+3.20%	+0.70pp
WALCL_contract_2.0_CURRENT_EXTREME	Y	-2.0	11	27.3%	+2.97%	+0.47pp

20a. WALCL: all horizons per band

WALCL_expand_0.3 | threshold=0.3 | direction=UP | bullish=True | events n=95

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	94	61.7%	+3.19%	-4.16%	+0.38%	+0.5%	-0.12pp	+0.38pp
3M (63d)	94	70.2%	+6.51%	-6.00%	+2.78%	+2.5%	+0.28pp	+0.57pp
6M (126d)	92	73.9%	+9.25%	-11.10%	+3.94%	+5.0%	-1.06pp	+0.65pp
9M (189d)	92	77.2%	+12.21%	-17.01%	+5.54%	+7.5%	-1.96pp	+0.11pp
12M (252d)	92	78.3%	+14.49%	-20.80%	+6.82%	+10.0%	-3.18pp	-0.31pp

WALCL_expand_0.5 | threshold=0.5 | direction=UP | bullish=True | events n=86

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	86	66.3%	+3.43%	-4.51%	+0.76%	+0.5%	+0.26pp	+0.90pp
3M (63d)	85	67.1%	+7.25%	-5.52%	+3.04%	+2.5%	+0.54pp	+0.43pp

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
6M (126d)	83	69.9%	+9.41%	-11.43%	+3.13%	+5.0%	-1.87pp	-0.32pp
9M (189d)	83	72.3%	+12.30%	-16.68%	+4.27%	+7.5%	-3.23pp	-1.64pp
12M (252d)	83	72.3%	+14.60%	-19.56%	+5.13%	+10.0%	-4.87pp	-3.13pp

WALCL_expand_0.8_CURRENT_RARE CURRENT**** | threshold=0.8 | direction=UP |
bullish=True | events n=76

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	76	67.1%	+3.23%	-4.05%	+0.84%	+0.5%	+0.34pp	+1.29pp
3M (63d)	76	68.4%	+6.48%	-7.33%	+2.12%	+2.5%	-0.38pp	+0.53pp
6M (126d)	75	70.7%	+9.24%	-14.15%	+2.38%	+5.0%	-2.62pp	+0.64pp
9M (189d)	75	72.0%	+11.62%	-17.23%	+3.54%	+7.5%	-3.96pp	+0.17pp
12M (252d)	75	73.3%	+14.31%	-20.88%	+4.92%	+10.0%	-5.08pp	-1.40pp

WALCL_expand_1.5 | threshold=1.5 | direction=UP | bullish=True | events n=52

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	52	65.4%	+3.53%	-4.13%	+0.88%	+0.5%	+0.38pp	+1.86pp
3M (63d)	52	71.2%	+6.18%	-8.35%	+1.99%	+2.5%	-0.51pp	+3.45pp
6M (126d)	51	64.7%	+9.39%	-12.65%	+1.61%	+5.0%	-3.39pp	+2.33pp
9M (189d)	51	66.7%	+11.79%	-15.27%	+2.77%	+7.5%	-4.73pp	+1.56pp
12M (252d)	51	70.6%	+14.32%	-21.20%	+3.87%	+10.0%	-6.13pp	-0.41pp

WALCL_expand_2.0_CURRENT_EXTREME CURRENT**** | threshold=2.0 | direction=UP |
bullish=True | events n=43

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	43	69.8%	+3.90%	-4.37%	+1.40%	+0.5%	+0.90pp	+2.07pp
3M (63d)	43	81.4%	+6.77%	-10.47%	+3.56%	+2.5%	+1.06pp	+5.73pp
6M (126d)	43	69.8%	+10.53%	-10.61%	+4.14%	+5.0%	-0.86pp	+2.24pp
9M (189d)	43	76.7%	+13.59%	-10.56%	+7.97%	+7.5%	+0.47pp	+1.56pp
12M (252d)	43	79.1%	+17.49%	-16.64%	+10.35%	+10.0%	+0.35pp	+3.37pp

WALCL_expand_3.0 | threshold=3.0 | direction=UP | bullish=True | events n=22

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	22	72.7%	+3.49%	-6.71%	+0.71%	+0.5%	+0.21pp	+5.58pp
3M (63d)	22	68.2%	+8.19%	-8.52%	+2.87%	+2.5%	+0.37pp	+10.09pp
6M (126d)	22	68.2%	+10.15%	-15.30%	+2.05%	+5.0%	-2.95pp	+8.71pp
9M (189d)	22	63.6%	+17.28%	-10.99%	+7.00%	+7.5%	-0.50pp	+12.99pp
12M (252d)	22	86.4%	+16.71%	-18.66%	+11.89%	+10.0%	+1.89pp	+22.22pp

WALCL_contract_0.3 | threshold=-0.3 | direction=DOWN | bullish=False | events n=89

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	89	31.5%	-3.18%	+3.34%	+1.29%	+0.5%	+0.79pp	+1.04pp
3M (63d)	89	25.8%	-6.72%	+6.20%	+2.86%	+2.5%	+0.36pp	+0.52pp
6M (126d)	88	22.7%	-10.52%	+8.88%	+4.47%	+5.0%	-0.53pp	-0.20pp
9M (189d)	87	24.1%	-13.27%	+12.47%	+6.26%	+7.5%	-1.24pp	-0.85pp
12M (252d)	84	20.2%	-21.10%	+15.05%	+7.73%	+10.0%	-2.27pp	-2.16pp

WALCL_contract_0.5 | threshold=-0.5 | direction=DOWN | bullish=False | events n=85

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	85	29.4%	-3.85%	+3.16%	+1.10%	+0.5%	+0.60pp	+0.67pp
3M (63d)	85	27.1%	-6.67%	+5.91%	+2.50%	+2.5%	+0.00pp	+0.46pp
6M (126d)	84	23.8%	-9.04%	+8.25%	+4.13%	+5.0%	-0.87pp	-0.06pp
9M (189d)	83	25.3%	-12.17%	+11.75%	+5.70%	+7.5%	-1.80pp	-1.85pp
12M (252d)	82	20.7%	-19.68%	+15.30%	+8.05%	+10.0%	-1.95pp	-0.93pp

WALCL_contract_0.8_CURRENT_RARE CURRENT**** | threshold=-0.8 | direction=DOWN | bullish=False | events n=66

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	66	36.4%	-3.35%	+3.43%	+0.96%	+0.5%	+0.46pp	+0.60pp
3M (63d)	66	30.3%	-6.40%	+6.35%	+2.49%	+2.5%	-0.01pp	+0.71pp
6M (126d)	66	25.8%	-10.51%	+8.23%	+3.41%	+5.0%	-1.59pp	+0.24pp
9M (189d)	66	21.2%	-15.71%	+11.56%	+5.78%	+7.5%	-1.72pp	-0.25pp
12M (252d)	66	21.2%	-19.37%	+15.33%	+7.97%	+10.0%	-2.03pp	-0.49pp

WALCL_contract_1.5 | threshold=-1.5 | direction=DOWN | bullish=False | events n=24

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	24	41.7%	-2.99%	+2.92%	+0.46%	+0.5%	-0.04pp	+0.00pp
3M (63d)	24	29.2%	-3.58%	+6.00%	+3.20%	+2.5%	+0.70pp	+2.18pp
6M (126d)	24	25.0%	-2.89%	+8.43%	+5.60%	+5.0%	+0.60pp	+2.08pp
9M (189d)	24	16.7%	-11.32%	+15.04%	+10.65%	+7.5%	+3.15pp	+6.04pp
12M (252d)	24	12.5%	-17.24%	+18.70%	+14.21%	+10.0%	+4.21pp	+6.07pp

WALCL_contract_2.0_CURRENT_EXTREME CURRENT**** | threshold=-2.0 | direction=DOWN
| bullish=False | events n=11

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	11	63.6%	-3.76%	+3.35%	-1.18%	+0.5%	-1.68pp	-1.43pp
3M (63d)	11	27.3%	-3.49%	+5.40%	+2.97%	+2.5%	+0.47pp	+1.07pp
6M (126d)	11	36.4%	-4.06%	+9.23%	+4.40%	+5.0%	-0.60pp	-3.91pp
9M (189d)	11	27.3%	-10.60%	+15.73%	+8.55%	+7.5%	+1.05pp	-0.15pp
12M (252d)	11	18.2%	-22.82%	+17.02%	+9.78%	+10.0%	-0.22pp	-6.75pp

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21. WTI: full sweep data

CONFIG RARE: {"pct_4wk": 6.0}

CONFIG EXTREME: {"pct_4wk": 10.0}

Primary horizon: 6M (126d)

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
WTI_down_3pct		-3	139	72.4%	+4.66%	-0.34pp
WTI_down_5pct		-5	121	67.8%	+4.28%	-0.72pp
WTI_down_6pct _CURRENT_RARE	Y	-6	116	67.9%	+4.35%	-0.65pp
WTI_down_8pct		-8	90	68.6%	+3.34%	-1.66pp
WTI_down_10pc t_CURRENT_EXTREME	Y	-10	73	68.6%	+3.52%	-1.48pp
WTI_down_15pc t		-15	31	73.3%	+5.16%	+0.16pp
WTI_up_5pct		5	142	30.9%	+3.22%	-1.78pp
	Y	6	134	31.5%	+3.01%	-1.99pp

Band	Current?	Threshold	n	Hit @ primary	PW expected	Excess
WTI_up_6pct_C URRENT_RARE						
WTI_up_8pct		8	119	27.8%	+3.47%	-1.53pp
WTI_up_10pct_C URRENT_EXTRE ME	Y	10	96	33.3%	+1.87%	-3.13pp
WTI_up_15pct		15	42	37.5%	+2.81%	-2.19pp

21a. WTI: all horizons per band

WTI_down_3pct | threshold=-3 | direction=DOWN | bullish=True | events n=139

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	137	66.4%	+3.76%	-3.87%	+1.20%	+0.5%	+0.70pp	+1.03pp
3M (63d)	136	71.3%	+6.42%	-6.84%	+2.62%	+2.5%	+0.12pp	+0.70pp
6M (126d)	134	72.4%	+10.04%	-9.44%	+4.66%	+5.0%	-0.34pp	+1.32pp
9M (189d)	132	78.8%	+13.05%	-11.61%	+7.82%	+7.5%	+0.32pp	+1.71pp
12M (252d)	129	80.6%	+15.66%	-16.12%	+9.50%	+10.0%	-0.50pp	+2.04pp

WTI_down_5pct | threshold=-5 | direction=DOWN | bullish=True | events n=121

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	119	63.9%	+3.64%	-4.61%	+0.66%	+0.5%	+0.16pp	+0.30pp
3M (63d)	118	68.6%	+6.59%	-6.45%	+2.50%	+2.5%	+0.00pp	-0.22pp
6M (126d)	118	67.8%	+10.39%	-8.59%	+4.28%	+5.0%	-0.72pp	-0.38pp
9M (189d)	116	79.3%	+12.95%	-13.57%	+7.46%	+7.5%	-0.04pp	+0.77pp
12M (252d)	114	79.8%	+15.57%	-17.32%	+8.94%	+10.0%	-1.06pp	+0.87pp

WTI_down_6pct_CURRENT_RARE CURRENT**** | threshold=-6 | direction=DOWN | bullish=True | events n=116

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	114	65.8%	+3.59%	-4.11%	+0.96%	+0.5%	+0.46pp	+0.43pp
3M (63d)	112	71.4%	+6.39%	-6.84%	+2.61%	+2.5%	+0.11pp	-0.82pp
6M (126d)	112	67.9%	+10.37%	-8.35%	+4.35%	+5.0%	-0.65pp	-0.98pp
9M (189d)	111	77.5%	+13.16%	-13.25%	+7.21%	+7.5%	-0.29pp	-0.14pp
12M (252d)	109	78.9%	+15.70%	-17.18%	+8.76%	+10.0%	-1.24pp	-0.30pp

WTI_down_8pct | threshold=-8 | direction=DOWN | bullish=True | events n=90

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	88	64.8%	+3.70%	-5.41%	+0.49%	+0.5%	-0.01pp	+0.80pp
3M (63d)	86	68.6%	+6.53%	-8.59%	+1.78%	+2.5%	-0.72pp	-0.21pp
6M (126d)	86	68.6%	+10.02%	-11.26%	+3.34%	+5.0%	-1.66pp	-0.92pp
9M (189d)	85	72.9%	+13.67%	-15.47%	+5.79%	+7.5%	-1.71pp	+0.59pp
12M (252d)	84	73.8%	+16.63%	-15.61%	+8.19%	+10.0%	-1.81pp	+1.45pp

WTI_down_10pct_CURRENT_EXTREME CURRENT**** | threshold=-10 | direction=DOWN | bullish=True | events n=73

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	71	63.4%	+3.56%	-5.47%	+0.25%	+0.5%	-0.25pp	+0.85pp
3M (63d)	70	65.7%	+6.58%	-9.53%	+1.06%	+2.5%	-1.44pp	+0.16pp
6M (126d)	70	68.6%	+9.73%	-10.01%	+3.52%	+5.0%	-1.48pp	+0.40pp
9M (189d)	70	75.7%	+13.85%	-13.38%	+7.24%	+7.5%	-0.26pp	+1.61pp
12M (252d)	69	76.8%	+16.74%	-11.04%	+10.30%	+10.0%	+0.30pp	+2.73pp

WTI_down_15pct | threshold=-15 | direction=DOWN | bullish=True | events n=31

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	30	76.7%	+3.40%	-6.07%	+1.19%	+0.5%	+0.69pp	+0.63pp
3M (63d)	30	63.3%	+7.11%	-7.44%	+1.78%	+2.5%	-0.72pp	-1.15pp
6M (126d)	30	73.3%	+10.56%	-9.67%	+5.16%	+5.0%	+0.16pp	+1.15pp
9M (189d)	30	70.0%	+14.84%	-13.03%	+6.48%	+7.5%	-1.02pp	-1.64pp
12M (252d)	30	73.3%	+17.60%	-14.11%	+9.14%	+10.0%	-0.86pp	+0.45pp

WTI_up_5pct | threshold=5 | direction=UP | bullish=False | events n=142

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	141	36.9%	-3.77%	+2.87%	+0.42%	+0.5%	-0.08pp	+0.44pp
3M (63d)	141	30.5%	-5.35%	+5.46%	+2.16%	+2.5%	-0.34pp	+0.42pp
6M (126d)	139	30.9%	-9.32%	+8.84%	+3.22%	+5.0%	-1.78pp	+0.86pp
9M (189d)	139	25.2%	-13.24%	+11.66%	+5.39%	+7.5%	-2.11pp	-0.32pp
12M (252d)	139	24.5%	-14.01%	+15.06%	+7.95%	+10.0%	-2.05pp	-1.05pp

WTI_up_6pct_CURRENT_RARE CURRENT**** | threshold=6 | direction=UP | bullish=False | events n=134

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	133	37.6%	-4.19%	+2.76%	+0.15%	+0.5%	-0.35pp	+0.24pp
3M (63d)	133	31.6%	-5.48%	+5.39%	+1.95%	+2.5%	-0.55pp	+0.59pp
6M (126d)	130	31.5%	-9.98%	+9.00%	+3.01%	+5.0%	-1.99pp	+1.14pp
9M (189d)	130	25.4%	-14.17%	+11.56%	+5.03%	+7.5%	-2.47pp	+0.04pp
12M (252d)	129	25.6%	-14.96%	+14.93%	+7.28%	+10.0%	-2.72pp	-0.30pp

WTI_up_8pct | threshold=8 | direction=UP | bullish=False | events n=119

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	118	42.4%	-3.81%	+3.10%	+0.17%	+0.5%	-0.33pp	-0.32pp
3M (63d)	118	30.5%	-5.54%	+5.65%	+2.24%	+2.5%	-0.26pp	+0.58pp
6M (126d)	115	27.8%	-10.33%	+8.79%	+3.47%	+5.0%	-1.53pp	+0.55pp
9M (189d)	115	27.0%	-13.85%	+11.86%	+4.93%	+7.5%	-2.57pp	-0.29pp
12M (252d)	114	25.4%	-16.16%	+14.99%	+7.07%	+10.0%	-2.93pp	-0.39pp

WTI_up_10pct_CURRENT_EXTREME CURRENT**** | threshold=10 | direction=UP | bullish=False | events n=96

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	95	43.2%	-3.60%	+3.12%	+0.22%	+0.5%	-0.28pp	-0.56pp
3M (63d)	95	35.8%	-5.79%	+5.93%	+1.73%	+2.5%	-0.77pp	+0.70pp
6M (126d)	93	33.3%	-12.42%	+9.01%	+1.87%	+5.0%	-3.13pp	+0.02pp
9M (189d)	93	29.0%	-17.49%	+11.96%	+3.41%	+7.5%	-4.09pp	-0.31pp
12M (252d)	92	30.4%	-19.13%	+15.55%	+4.99%	+10.0%	-5.01pp	-1.13pp

WTI_up_15pct | threshold=15 | direction=UP | bullish=False | events n=42

Horizon	n	Hit rate	Avg win	Avg loss	PW expected	Benchmark	Excess	Hostile excess
1M (21d)	41	39.0%	-4.10%	+3.07%	+0.27%	+0.5%	-0.23pp	-0.12pp
3M (63d)	41	31.7%	-7.45%	+6.74%	+2.24%	+2.5%	-0.26pp	+0.19pp
6M (126d)	40	37.5%	-10.91%	+11.04%	+2.81%	+5.0%	-2.19pp	-1.42pp
9M (189d)	40	32.5%	-17.24%	+14.71%	+4.33%	+7.5%	-3.17pp	+0.50pp
12M (252d)	39	30.8%	-19.41%	+17.79%	+6.35%	+10.0%	-3.65pp	+2.87pp

22. Combo B (Capitulation): gate sweep detail

Combo: B

Validated Horizon: spx_3m

Direction: bullish

Current Gates: {'all_required': ['VIX', 'HY', 'CFTC'], 'vix_min': 25, 'hy_bps_min': 400, 'cftc_max_pctile': 15}

Note: Leg replay on daily_readings (0 ACTIVE in combo_fires; 89 WATCH)

Run Date: 2026-06-15

Validated horizon: 3M (63d) | **Benchmark:** +2.5%

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CB_VIX_20	vix_min	20	3	0	n/a	n/a	n/a	n/a	n/a
CB_VIX_25	vix_min	25	3	0	n/a	n/a	n/a	n/a	n/a
CB_VIX_30	vix_min	30	3	0	n/a	n/a	n/a	n/a	n/a
CB_HY_350	hy_bps_min	350	3	0	n/a	n/a	n/a	n/a	n/a
CB_HY_400	hy_bps_min	400	3	0	n/a	n/a	n/a	n/a	n/a
CB_HY_450	hy_bps_min	450	3	0	n/a	n/a	n/a	n/a	n/a
CB_CFTC_10	cftc_max_pctile	10	3	0	n/a	n/a	n/a	n/a	n/a
CB_CFTC_15	cftc_max_pctile	15	3	0	n/a	n/a	n/a	n/a	n/a
CB_CFTC_20	cftc_max_pctile	20	3	0	n/a	n/a	n/a	n/a	n/a
CB_2of3_legs	legs_required	2	2	12	75.0%	+10.41%	-0.99%	+7.56%	+5.06pp

23. Combo F (Recovery): SPX threshold sweep detail

Combo: F

Validated Horizon: spx_6m

Direction: bullish

Current Gates: {'spx_50wma_reclaim_weekly_pct': 3.0, 'cftc_max_pctile': 50,

'active_weeks': 26}

Run Date: 2026-06-15

Validated horizon: 6M (126d) | Benchmark: +5.0%

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CF_SPX_1pct	spx_50wma_reclaim_weekly_pct	1.0	?	46	80.0%	+10.53%	-5.05%	+7.41%	+2.41pp
CF_SPX_2pct	spx_50wma_reclaim_weekly_pct	2.0	?	46	82.2%	+10.11%	-5.87%	+7.27%	+2.27pp
CF_SPX_3pct	spx_50wma_reclaim_weekly_pct	3.0	?	43	85.7%	+10.22%	-4.25%	+8.15%	+3.15pp
CF_SPX_5pct	spx_50wma_reclaim_weekly_pct	5.0	?	42	85.4%	+10.53%	-3.35%	+8.50%	+3.50pp

24. Combo E (Valuation Extreme): gate sweep detail

Combo: E

Validated Horizon: spx_12m

Direction: bearish

Current Gates: {'min_of_three': 2, 'cape_min': 28, 'nfc_i_easy_max': -0.3, 'cftc_min_pctile': 80}

Run Date: 2026-06-15

Validated horizon: 12M (252d) | Benchmark: +10.0%

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CE_CAPE_25	cape_min	25	?	18	5.6%	-1.07%	+18.61%	+17.51%	+7.51pp
CE_CAPE_28	cape_min	28	?	22	9.1%	-2.37%	+17.79%	+15.96%	+5.96pp
CE_CAPE_30	cape_min	30	?	31	9.7%	-4.22%	+16.77%	+14.74%	+4.74pp
CE_CAPE_32	cape_min	32	?	30	6.7%	-2.37%	+15.53%	+14.33%	+4.33pp

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CE_NFCI_0.2	nfc_i_easy_max	-0.2	?	23	8.7%	-2.37%	+16.99%	+15.31%	+5.31pp
CE_NFCI_0.3	nfc_i_easy_max	-0.3	?	22	9.1%	-2.37%	+17.79%	+15.96%	+5.96pp
CE_NFCI_0.4	nfc_i_easy_max	-0.4	?	21	9.5%	-1.98%	+17.35%	+15.51%	+5.51pp
CE_CFTC_75	cftc_min_pctile	75	?	27	11.1%	-2.05%	+17.22%	+15.07%	+5.07pp
CE_CFTC_80	cftc_min_pctile	80	?	22	9.1%	-2.37%	+17.79%	+15.96%	+5.96pp
CE_CFTC_85	cftc_min_pctile	85	?	19	10.5%	-1.51%	+16.69%	+14.78%	+4.78pp

25. Combo D (FOMO Top): gate sweep detail

Combo: D

Validated Horizon: spx_5d

Direction: bearish

Current Gates: {'vxts_min': 1.1, 'cftc_min_pctile': 85, 'vix_max': 18}

Run Date: 2026-06-15

Validated horizon: 5D (5d) | **Benchmark:** +0.5%

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CD_VXTS_1.05	vxts_min	1.05	3	32	37.5%	-1.25%	+1.18%	+0.27%	-0.23pp
CD_VXTS_1.1	vxts_min	1.1	3	31	41.9%	-1.18%	+1.23%	+0.22%	-0.28pp
CD_VXTS_1.15	vxts_min	1.15	3	25	40.0%	-1.39%	+0.96%	+0.02%	-0.48pp
CD_CFTC_80	cftc_min_pctile	80	3	33	39.4%	-0.95%	+1.26%	+0.39%	-0.11pp
CD_CFTC_85	cftc_min_pctile	85	3	31	41.9%	-1.18%	+1.23%	+0.22%	-0.28pp
CD_CFTC_90	cftc_min_pctile	90	3	25	40.0%	-1.43%	+1.23%	+0.17%	-0.33pp
CD_VIX_15	vix_max	15	3	24	45.8%	-1.32%	+1.17%	+0.03%	-0.47pp
CD_VIX_18	vix_max	18	3	31	41.9%	-1.18%	+1.23%	+0.22%	-0.28pp
CD_VIX_20	vix_max	20	3	38	42.1%	-1.56%	+1.40%	+0.16%	-0.34pp

Test ID	Parameter	Value	Legs req	n	Hit rate	Avg win	Avg loss	PW expected	Excess
CD_2of3_legs	legs_required	2	2	87	33.7%	-1.76%	+1.30%	+0.27%	-0.23pp
